

Informed Trading and Price Discovery before Corporate Events

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Abstract

Stock prices incorporate less “news” before negative events than positive events. Further, we find evidence that informed agents use less price aggressive (limit) orders before negative events and more price aggressive (market) orders before positive events (“buy-sell asymmetry”). Motivated by these patterns, we model the execution risk that informed agents impose on each other and relate the asymmetry to costly short selling. When investor base is narrow, security borrowing is difficult, or the magnitude of the event is small, buy-sell asymmetry is pronounced and price discovery before negative events is lower. Overall, informed agents' strategies influence the process of price formation in financial markets, as predicted by theory.

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Abstract

Stock prices incorporate less “news” before negative events than positive events. Further, we find evidence that informed agents use less price aggressive (limit) orders before negative events and more price aggressive (market) orders before positive events (“buy-sell asymmetry”). Motivated by these patterns, we model the execution risk that informed agents impose on each other and relate the asymmetry to costly short selling. When investor base is narrow, security borrowing is difficult, or the magnitude of the event is small, buy-sell asymmetry is pronounced and price discovery before negative events is lower. Overall, informed agents' strategies influence the process of price formation in financial markets, as predicted by theory.

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JEL classification: G11, G12, G14, G18.

1. Introduction

Informed traders play a central role in price formation in financial markets. In microstructure models (see Kyle, 1985; Glosten and Milgrom, 1985), informed traders receive a private signal about the security's value and build positions before the information is widely available. The market participants observe the imbalance created by the informed order flow and move prices in the direction of the signal. Thus, the theoretical literature predicts a direct link between the strategies of informed traders and the informational efficiency of prices before news releases.

In this study, we present an extension of the theory describing informed trader's strategies, and present relevant empirical evidence on the most basic questions about informed trading: First, how do informed traders build positions, and what explains their strategies? Second, how do their decisions to use aggressive or passive strategies affect the process through which private information is incorporated into security prices? And third, how do their strategies affect the speed of learning before news releases? Understanding the trading behavior of informed agents is central both to validating the well-developed theoretical literature that relates trading strategies to price discovery, and to designing better surveillance systems that monitor and detect insider trading activity.¹

Despite the importance of understanding how informed agents build positions, data limitations have left unanswered many important questions about their strategies. In the days leading to corporate announcements, such as those related to mergers and acquisitions (M&As) and earnings releases, the empirical literature documents widespread evidence of both insider trading activity and significant stock price movements.² Many publicly available data sources, such as NYSE's Trade and Quote (TAQ) database, report all the transactions in a market but do not identify the specific trades of informed agents. Some data on transactions are available from regulatory Form 4 filed by corporate insiders with the U.S.

¹ In this study, the term informed trader refers to any person with access to valuable, non public information before a corporate announcement. An important category of informed traders are corporate insiders, such as board members, directors, and employees. Well-connected market participants such as bankers, analysts, lawyers, hedge fund managers, and friends of corporate executives, also could have access to information about pending news releases.

² See, for example, Keown and Pinkerton (1981) on price reaction, Meulbroek (1992) on insider trading, and Bodnaruk, Massa, and Simonov (2009) and Jegadeesh and Tang (2010) on trading by affiliated investment banks and institutional investors before corporate announcements.

Securities and Exchange Commission (SEC) (see Seyhun, 1986), or from the SEC's case files of defendants formally charged with insider trading (see Meulbroek, 1992); however, these datasets do not contain order level information that is necessary to characterize the strategies of informed agents.

We examine a dataset provided by the Euronext-Paris exchange that contains detailed information on all orders submitted for all stocks; however the Euronext data do not directly identify the orders of informed agents. To identify informed order flow, we employ a research design based on the methodology from Chae (2005), Graham, Koski, and Loewenstein (2006), and Sarkar and Schwartz (2009). These studies show that uninformed traders decrease their participation in periods leading to news releases with timing that is known in advance, such as earnings announcements (see Lee, Mucklow, and Ready, 1993). In contrast, uninformed traders cannot anticipate *unscheduled* news events with timing that is a complete surprise, but informed traders can, if their information concerns the event.³ We therefore examine a sample of *unscheduled* corporate announcements related to M&As, Seasoned Equity Offerings (SEOs), stock repurchases, dividend initiations, and dividend terminations for Euronext-Paris stocks in 2003.⁴ Following the prior literature, we attribute the change in trading activity observed before an unscheduled announcement relative to a control period for the same firm to the actions of informed agents. By holding trading activity in each firm as its own benchmark, our research design reduces the influence of time-invariant firm attributes on the variation in trading strategies across firms.

We present a model of the informed agent's choice of market versus limit orders. The intuition for the model is as follows. Informed agents face a tradeoff between transacting with certainty at a current bid or offer price by placing a market order versus managing execution risk in an attempt to get a better price by placing a limit order. In addition to paying the bid-ask spread, market orders tip off market participants about the private signal and increase price impact cost. When other informed agents receive

³ A WSJ article dated 06/06/2013 reports on the increase in trading activity observed before Smithfield's acquisition announcement, saying "*When multiple bidders vie for a company, it isn't unusual for hundreds of people to know about the possible deal before it surfaces – including employees of banks, law firms, and other outside advisors, not to mention the people inside the companies themselves.*"

⁴ More recent order-level data obtained from NYSE-Euronext have important limitations, which we describe in Section 2.1.

the same signal and trade on the same side of the market, the execution risk of a limit order strategy is particularly high. Our model predicts that informed agents use limit orders when there is sufficient uncertainty about the presence of other informed agents, and use market orders if they are certain that other informed agents are present. These predictions follow Kaniel and Liu's (2006) theoretical work showing that informed agents might use limit orders if the mass of informed agents is sufficiently low.

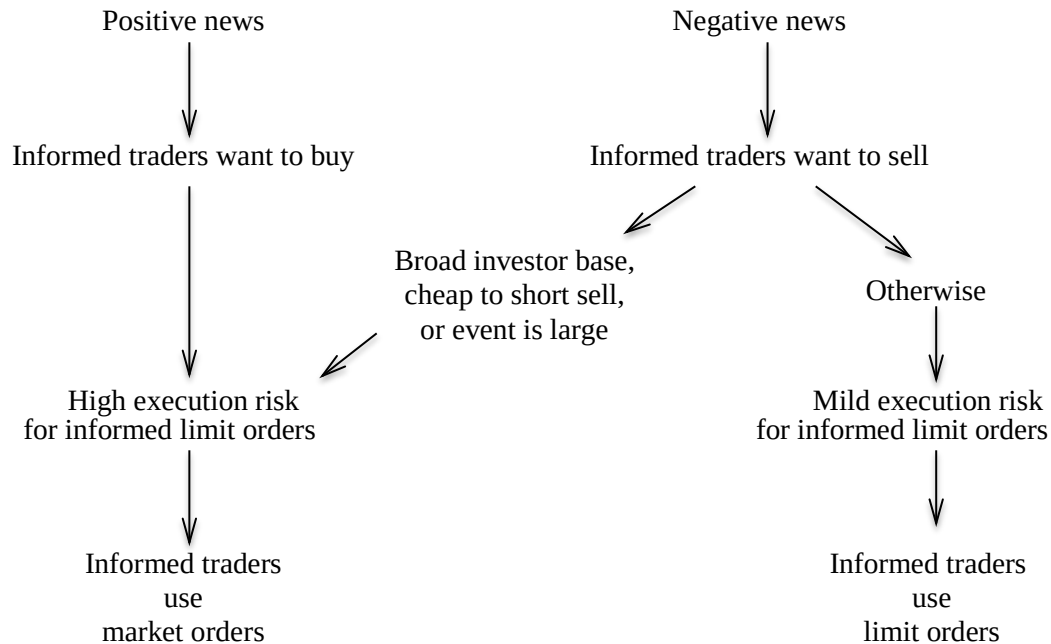
Informed agents face less competition when the nature of private information conveys a *decrease* in stock price. This is because informed agents are less likely to sell stocks with unfavorable information if they do not already own the stock.⁵ Consistent with this prediction, we report a novel empirical regularity of an asymmetry in informed agents' order submission strategies before positive and negative events. Specifically we document an *increase* in aggressively priced buy orders before positive events but a *decrease* in aggressively priced sell orders before negative events (henceforth "buy-sell asymmetry"). As in Diamond and Verrecchia (1987), our model relies on short sale constraints to explain the buy-sell asymmetry observed in the data. However, in our model, the asymmetry emerges not only because some informed sellers decide to abstain, but also because informed sellers become liquidity providers.

Our model yields cross-sectional predictions in buy-sell asymmetry based on anticipated competition among informed sellers. Informed sellers can be one of two types; the first type already owns the stock while the second type does not. The probability that the informed seller is of the first type increases with broadness of investor base. When investor base is narrow, when the cost of borrowing shares is large, and when the magnitude of news release is small such that the potential gains do not justify the borrowing costs, a limit order equilibrium emerges in which the first type of agent uses limit orders and the second type abstains from trade. On the other hand, when investor base is broad, when borrowing costs are small, or when the magnitude of news release is large, the second type of agent borrows the shares, and both types trade. The competition among informed sellers increases the execution

⁵ Corporate insiders face more constraints when they trade on bad news than on good news (see Marin and Olivier, 2008). Insiders in many markets are prohibited from selling short their own stock (e.g., Section 16 of the U.S. Securities Exchange Act of 1934), or sell stock holdings that are part of a compensation contract below a certain threshold. Informed sellers who do not own the stock incur the cost of borrowing shares. When borrowing costs are high, informed sellers anticipate less competition from other agents prior to information events.

risk of a limit order strategy and leads to usage of market orders. When the private information conveys an *increase* in stock price, informed buyers always anticipate competition and use market orders. Fig. 1 shows how anticipated competition among informed agents can lead to buy-sell asymmetry before corporate events.

Fig. 1: Execution risk and order aggressiveness



Motivated by prior work, our empirical measures of short sale constraints include stock index membership, availability of exchange-listed stock options, and eligibility for Euronext’s Deferred Settlement Service. Index stocks have broad investor base, active participation by institutions, and lower borrowing cost (see D’Avolio, 2002; Nagel, 2005). Stock options allow informed sellers to establish equivalent positions in short constrained stocks at a lower cost (see Battalio and Schultz, 2011; Hu, 2014). Euronext’s Deferred Settlement Service (called “SRD”) facility allows short positions in eligible stocks to be established more easily (see Foucault, Sraer, and Thesmar, 2011).

We find strong empirical support for the model’s predictions. In scenarios where informed traders anticipate competition (i.e., index constituent stocks, stocks with listed options, SRD-eligible stocks, or

large news announcements), we observe an increase in price aggressiveness for buy orders before positive events *and* for sell orders before negative events. In contrast, when short selling is costly (i.e., non-index stocks, stocks without listed options, or SRD-ineligible stocks) or when news is small, we observe an asymmetry in the mix of order flow before positive and negative events. Specifically, the price aggressiveness of buy orders increases before positive events while the price aggressiveness of sell orders decreases before negative events. Results are robust to different methods for classifying events as positive and negative, as well as alternative model specifications, including panel regressions.

Further, we report the novel empirical result that stock prices incorporate less “news” before negative events than positive events. Following Biais, Hillion, and Spatt (1999), we estimate the efficiency of price discovery before corporate events using “unbiasedness regressions”, where we regress the close-to-close return during Days $[-6,+1]$ on the return in the interval $[-6, I]$ where I represents half-hour snapshots in the interval $[-5,-1]$. Barclay and Hendershott (2003) interpret the slope of the unbiasedness regressions as the signal:noise ratio. In our research design, as the slope moves closer to “one”, the interpretation is that the security price before the event reflects the security price after the event with increasing precision. For positive and negative events, the slope of the unbiasedness regression at the beginning of the pre-event window is not statistically different from zero. As the event gets closer (i.e., Day $[-1]$), the slope for positive events is not significantly different from one while for negative events, the slope is lower (between zero and one) and significantly different from one. Additionally, consistent with higher price discovery before positive events, we estimate large opportunity costs when limit orders go unfilled before positive events but not before negative events, indicating that employing limit orders before positive events is a costly strategy.

Results based on subsamples suggest that informed traders’ strategies affect the informational efficiency of prices before corporate events. When informed traders employ less price aggressive orders (i.e., negative events in non-index stocks, SRD-ineligible stocks, and stocks without listed options), the slope of the unbiasedness regression towards the end of the pre-event window is between zero and one, and statistically different from one. When informed traders employ more price aggressive orders (i.e.,

negative events with no short sale constraints and all positive events), the slope of the unbiasedness regression is positive and not statistically different from one. Collectively, our results provide support for the well-established theoretical literature that links informed traders' strategies to level of competition, and further, to the process of price formation in financial markets (Holden and Subrahmanyam, 1992).

In the Diamond and Verrecchia (1987) and also Saar (2001) framework, informed traders trade upon arrival; in other words, they do not employ limit orders. In fact, most theoretical work posits that informed agents exclusively use market orders to exploit their information advantage.⁶ Novel exceptions are Kumar and Seppi (1994), Chakravarty and Holden (1995), Kaniel and Liu (2006), Goettler, Parlour, and Rajan (2009), Boulatov and George (2013), and Rosu (2014). In these models, informed traders do find it optimal under certain conditions to submit limit orders. However, none of these theoretical papers relate strategies of informed agents to the direction of the private signal, and further, to asymmetry in competition that is introduced by costly short selling.

In a recent work, Collin-Dufresne and Fos (2015) study a sample of trades reported in the Schedule 13D filings by activist investors. The study shows that activist investors time their trades when liquidity is high and use limit orders when public disclosure of their positions is not imminent. These results support Kaniel and Liu's (2006) prediction that informed traders with long-lived information use limit orders.⁷

Our study points to an unintended consequence of the widespread ban on short selling by regulators around the world in response to the 2007-09 financial crisis. Beber and Pagano (2013) find that the short sale ban lowers the information efficiency of prices, particularly surrounding events with negative information. Because the short sale ban reduces competition among sellers, our model predicts

⁶ See Kyle (1985) and Glosten and Milgrom (1985) for studies of strategic trading in a dealer setting and Glosten (1994), Rock (1996), Seppi (1997) and Back and Baruch (2013) for strategic trading in a limit order book setting.

⁷ Evidence on limit order usage in Collin-Dufresne and Fos (2015) is indirect because Schedule 13D filings do not require investors to disclose what type of orders they use. The study matches transaction data disclosed in Schedule 13D filings with Trade and Quote (TAQ) data and uses Lee and Ready (1991) algorithm to infer whether activist investors use limit orders. Other experimental and empirical papers also suggest that informed traders use limit orders. See Barclay, Hendershott, and McCormick (2003); Bloomfield, O'Hara, and Saar (2005, 2015); Anand, Chakravarty, and Martell (2005), Hautsch and Huang (2012), Zhang (2013), among others.

that informed sellers who already own the stock use less aggressively priced orders, which impedes the flow of negative private information into prices. Thus our study offers a specific mechanism by which short sale constraints impedes the price discovery process, as shown by Boehmer and Wu (2013).

The rest of the paper is organized as follows. Section 2 describes the sample, the data sources, and summary statistics. In Section 3, we present evidence of buy-sell asymmetry in order price aggressiveness, price discovery, and the opportunity cost of non-execution of limit orders before positive and negative events. In Section 4, we present a model that explains the results and provides new testable predictions. We test the model's cross-sectional predictions in Section 5 and present robustness analyses in Section 6. Section 7 presents the study's conclusions.

2. Sample and data

We examine the Euronext-Paris, Base de Donnees de Marche (BDM) database for the year 2003. The BDM database contains detailed information on the characteristics of all orders submitted for all stocks listed on Euronext-Paris. This includes the stock symbol; the date and time of order submission; whether the order is a buy or a sell; the total size of the order (in shares); the displayed size (in shares); an order type indicator for identifying market or limit orders; a limit price in the case of a limit order; and instructions on when the order will expire.

We examine the 2003 sample period because more recent order-level data purchased from the Euronext market have important inaccuracies. In particular, orders that never get executed, or orders with a hidden size that are partly executed, do not get reported to the database. The omission affects the accuracy of the reconstructed limit order book, the analysis of order submission strategies, and the construction of control variables used in some specifications.

The growth in the alternative trading venues in the last decade has caused equity markets in the United States and Europe to be highly fragmented. In a fragmented market, an analysis of the informed traders' strategies must consider the choice of the trading venue to execute an order, which introduces a layer of complexity due to the inter-market priority rules and the make-take pricing arrangement at each

trading venue. During our sample period, the vast majority of trading activity (over 90%) in French stocks was observed on Euronext-Paris. Thus the market structure in 2003 provides a simple laboratory to test theoretical predictions on informed trader strategies.

The Euronext database does not provide information on trader identity. To identify informed order flow, we examine trading activity before unscheduled corporate events. The timing of information release for scheduled events, such as earning releases, is publicly available in advance of announcement. Chae (2005), Graham, Koski, and Loewenstein (2006), and Sarkar and Schwartz (2009) show that, although traders do not know in advance the *information* contained in scheduled events, those traders with some discretion on timing of trades alter activity prior to these events. For example, Lee, Mucklow, and Ready (1993) find that market makers widen bid-ask spreads and lower inside depth before earnings releases. In comparison, informed agents alone are aware of the pending news release for unscheduled, or “surprise” events, where the timing of the information release is not known publicly in advance. Sarkar and Schwartz (2009) show that trading activity before unscheduled events is characterized by one-sided markets, which is consistent with the presence of informed agents. Following the prior literature, we attribute the abnormal trading activity that is observed before unscheduled events to informed agents.

We identify unscheduled events using the Global SDC database compiled by Thomson Financial Securities Data and the AMADEUS database provided by Bureau van Dijk. We focus on five types of unscheduled events: M&As, SEOs, repurchases, dividend initiations, and dividend terminations. We use Bloomberg and Factiva search engines to identify the date of the first news announcement about the event (Day [0]). We eliminate Euronext-Paris stocks that switch from continuous trading to call auctions (or vice-versa) or were de-listed from Euronext during Days [-30,+10] surrounding the event. The final sample consists of 101 unscheduled corporate events for 95 unique stocks.

In microstructure models, informed agents build positions in the direction of private signal before the information is widely available. We therefore classify events as positive and negative news based on the announcement (Days [0,+1]) cumulative abnormal returns (CAR), where CAC40 Index return serves

as the benchmark.⁸ In Table 1, Panel A, the 58 positive events have a mean (median) Days [0,+1] CAR of 4.84% (2.74%) and the 43 negative events have a mean (median) Days [0,+1] CAR of -4.53% (-2.57%). The largest CAR are observed for M&As followed by SEOs. We observe positive and negative news for all event types with the exception of dividend termination.⁹

In Table 1, Panel B, we examine the stock's price movements over a five-day window before the event. For all event types, the magnitude of Days [-5,+1] CAR in Panel B is larger than Days [0,+1] CAR in Panel A, implying the stock price moves in the direction of news release before public announcement. Further, the results in Panel C suggest that the stock price incorporates more information before positive events than negative events. Specifically, for positive events, the Days [0,+1] CAR accounts for 44.83% of the Days [-5,+1] CAR, implying that the stock price reflects 55.17% of the “news” before positive events. In comparison, the stock price reflects 26.48% of the “news” before negative events.

Table 2 presents the descriptive statistics on the sample and the order flow before news releases. Results in Panel A suggest that the positive event sample has larger market capitalization, higher stock price, lower return volatility, and smaller bid-ask spread than the negative event sample. The median percentage bid-ask spread is approximately 0.80% indicating that the benefit of using limit orders instead of market orders is economically significant.

For each unscheduled event, Table 2, Panel B presents descriptive statistics for buy orders before positive events and sell orders before negative events. Statistics based on daily averages are reported for event Days [-5,-1], control Days [-30,-10], and tests of difference between event and control days.¹⁰ We report the total number of orders (marketable as well as non-marketable), the order size, the percentage of marketable to non-marketable orders (henceforth, market:limit ratio) and the percentage of limit orders

⁸ Robustness analysis based on alternative classification schemes are presented in Section 6.

⁹ Asquith and Mullins (1986) and Graham et al. (2006) find negative announcement returns for 28% and 36%, respectively, of their dividend initiation sample. The former study notes that “for these firms investors are anticipating the initiation of dividends and were disappointed by the amount of the initial dividend.”

¹⁰ For a sample of French M&A announcements, Aktas et al. (2007) report that trading volume spikes in the five days preceding a merger announcement and that trading volume and bid-ask spreads increase for both acquirers and targets in the Days [-65,-6] before announcement. Informed trading in the control period lowers the statistical power of our research design to find empirical support for the model. We minimize the impact of leakage before the news release by deleting all events with pre-event “rumors” reported in press articles based on Factiva search.

with a hidden size. A higher (lower) market:limit ratio on event days relative to control days suggests that informed traders use more (less) aggressively priced orders to build positions before events.

The results suggest that informed trades are active before unscheduled events. For both positive and negative events, the daily number of orders and the order size are higher on event days relative to control days. The market:limit ratio increases before positive events (significant at the 1% level) and the ratio decreases marginally (significant at the 10% level) before negative events, suggesting that informed agents use a different mix of order flow based on the direction of the signal. We find that the percentage of limit orders with a hidden size is similar on event and control days.

3. Analysis of price aggressiveness, price discovery, and opportunity cost of non-execution

Why do stock prices incorporate more information before positive events than negative events? Is price discovery influenced by the informed agent's choice of active versus passive strategies? We further explore these questions by modeling the order submission strategy in a multivariate regression framework that controls for order attributes and the market conditions at the time of arrival of an order.

3.1. Methodology

The multivariate regressions are estimated on an *event-by-event* basis using all orders submitted on event Days [-5,+1] and control Days [-30,-10]. The control period provides a benchmark measure of the order submission process for the firm, absent the event. Using the control period to estimate the shift in order flow before the event reduces the likelihood that the event effect is explained by time-invariant firm characteristics that influence the order submission process on non-event days.

The null hypothesis is that the event has no impact on order flow, or that the event effect is zero. The event's impact is captured by *day* coefficients that measure the change in dependent variable on event days relative to control days. Extensions of the original Kyle model generate different predictions about the trading intensity of informed agents before news releases (e.g., Kyle, 1985; Back, Cao, and Willard, 2000; Baruch, 2002; Sastry, Smith, and Thompson, 2014). Since theory does not predict that trading intensity is uniform, we estimate separate coefficients for each day in the event period (Days [-5,-1]),

which imposes fewer constraints by allowing individual day effects to differ before news releases. Since informed agents can be active on any of the five days before the event, we calculate the combined effect of informed trading with a *cumulative* measure that sums up the individual day coefficients. When interpreting the results, we focus on both the individual day effects and the cumulative effect, and the associated t-statistics. In Section 6, we demonstrate robustness with a specification that considers a single coefficient to capture the event's impact on Days [-5,-1].

We present results that are aggregated across events using the Bayesian framework attributable to DuMouchel (1994) and implemented by Bessembinder, Panayides, and Venkataraman (2009). Similar to Harris and Piwowar (2006), the method assigns larger weight in the cross-sectional aggregation to those events whose parameters are estimated more precisely. Further, the method allows variation in the true parameter estimates across events, thus accommodating the theoretical prediction that informed trader strategies are influenced by firm characteristics, such as short sale constraints. Details of the aggregation methodology are provided in Appendix A. Although the events are not clustered in calendar time and thus viewed as cross-sectionally independent, we present robustness analyses based on alternate model specifications, including panel regressions, in Section 6.

3.2. Price aggressiveness of informed traders before positive and negative events

We examine the informed agents' preference of market versus limit orders based on a regression analysis of order price aggressiveness. The regression specification accounts for the impact of market conditions on an order O_{it} that arrives at time of submission 't' for event 'i', as follows:

$$\begin{aligned} PriceAggressive_{it} = & \gamma_0 + \gamma_1 DayMinus5_{it} + \gamma_2 DayMinus4_{it} + \gamma_3 DayMinus3_{it} + \gamma_4 DayMinus2_{it} + \\ & \gamma_5 DayMinus1_{it} + \gamma_6 Day0\&Plus1_{it} + \gamma_7 DayPlus2_{it} + \\ & \gamma_8 OrderExposure_{it} + \gamma_9 TotalOrderSize_{it} + \gamma_{10} Spread_{it} + \gamma_{11} DepthSame_{it} + \gamma_{12} DepthOpp_{it} + \\ & \gamma_{13} BookOrderImbalance_{it} + \gamma_{14} Volatility_{it} + \gamma_{15} WaitTime_{it} + \gamma_{16} TradeFreqHour_{it} + \\ & \gamma_{17} PriceAggressive_{it-1} + \gamma_{18} DisplayedOrderSize_{it-1} + \gamma_{19} TradeSize_{it-1} + \gamma_{20} HiddenOppSide_{it-1} + \\ & \gamma_{21} FirstHour_{it} + \gamma_{22} LastHour_{it} + \gamma_{23} Mkt.Volatility_{it} + \gamma_{24} Ind.Volatility_{it} \end{aligned} \quad (1)$$

Following Biais, Hillion and Spatt (1995) and Bessembinder, Panayides, and Venkataraman (2009), *PriceAggressive* is an ordinal variable that takes the value of 1 when O_{it} is classified in the most

aggressive category and takes a value of 7 when O_{it} is classified in the least aggressive category. The first four categories represent orders that demand liquidity from the limit order book and the last three categories represent orders that supply liquidity to the book.^{11, 12}

To render the results more comparable across stocks, we normalize the continuous (non-dummy) variables by dividing the actual observation by the median for the stock on control days. *OrderExposure* equals one if the order O_{it} that arrives at time 't' has a hidden size, and equals zero otherwise. *TotalOrderSize* is total (displayed plus hidden) size of the order O_{it} divided by median order size during the control period.

The variables describing market conditions at time of order submission are defined as follows. *Spread* is the percentage bid-ask spread at time t divided by the median percentage spread during the control period. *DepthSame* is the displayed depth at the best bid (ask) when O_{it} is a buy (sell) order divided by the median depth during the control period. *DepthOpp* is the displayed depth at the best ask (bid) when O_{it} is a buy (sell) order divided by the median depth during the control period. *BookOrderImbalance* is the percentage difference between displayed liquidity in the best five prices on the same side versus the opposite side of order O_{it} in the limit order book divided by the median percentage difference during the control period. *Volatility* is the standard deviation of quote midpoint returns over the preceding hour divided by the median volatility during the control period. *WaitTime* is

¹¹ Specifically, the most aggressive orders (category 1) represents buy (sell) orders with order size greater than those displayed in the inside ask (bid) and with instructions to walk up (down) the book until the order is fully executed. Category 2 represents buy (sell) orders with order size greater than those displayed in the inside ask (bid) and with instructions to walk up (down) the book, but the order specifies a limit price such that the order is not expected to execute fully based on displayed book. Category 3 represents buy (sell) orders with the limit price equal to the inside ask (bid) and with order sizes greater than those displayed in the inside ask (bid). Orders in categories 2 and 3 could execute fully due to hidden liquidity but could also clear the book and convert to a standing limit order. Category 4 represents buy (sell) orders with the limit price equal to the inside ask (bid) and with order size less than those displayed in the inside ask (bid). These orders are expected to immediately execute in full. Category 5 represents orders with limit prices that lie within the inside bid and ask prices. Category 6 represents buy (sell) orders with limit price equal to the inside bid (ask). Finally, Category 7 represents buy (sell) orders with limit price less (greater) than the inside bid (ask).

¹² Following Liu and Agresti (2005) and Gelman & Hill (2007), we select a linear specification over a non-linear specification (ordered probit) because the dependent variable represents a large number of price aggressiveness (seven) categories. When fitting a proportional odds model, there is little gain in efficiency when using more than 4 levels of the category variable over an OLS. Further, a linear regression specification allows an easy estimation of the economic significance of day-indicator variables and the cumulative measure.

the average elapsed time between the prior three order arrivals on the same side as the order O_{it} divided by the median time during the control period. *TradeFreqHour* is the number of transactions during the previous hour of trading divided by the median number of transactions for the previous hour of trading during the control period. We also include control variables based on recent transactions observed before submission of the order O_{it} . Specifically, *TradesSize* is the size of the most recent transaction divided by the median trade size during the control period. *HiddenOppSide* is the size of the hidden depth at the best ask (bid) revealed in the prior transaction when O_{it} is a buy (sell) order divided by the median hidden depth (of the opposite side) revealed in control period. *DisplayedOrderSize* is the exposed size of the previous order (O_{it-1}) divided by median exposed size of all orders submitted during the control period.

Finally, we include control variables that capture industry volatility, market volatility, and time-of-the-day effects. Specifically, *FirstHour* (*LastHour*) is an indicator variable that equals one for orders submitted in the first (last) hour of the trading day, and equals zero otherwise. *Ind.Volatility* is the return volatility of the portfolio of all Euronext-Paris stocks belonging to the same industry while *Mkt.Volatility* is the return volatility of the CAC40 Index in the prior hour of trading, both divided by the median volatility numbers during the control period.¹³ The latter two variables help account for commonality in economic fundamentals (see Chordia, Roll, and Subrahmanyam, 2000).

Columns (1) and (2) of Table 3 report the coefficients of the price aggressiveness regression along with corresponding t-statistics, estimated on an event-by-event basis and then aggregated across events using the Bayesian Aggregation described in Appendix A. The coefficients on the control variables are consistent with those reported in the prior studies. Specifically, traders submit less aggressively priced orders (i.e., choose limit orders over market orders) when (a) the inside bid-ask spread is wide, (b) same side book depth is thin, which signals less competition (c) opposite side book depth is deep, or the last trade does not reveal the presence of hidden orders, both of which signal active counterparties, (d) volatility is high, consistent with a volatility capture strategy (see Handa and Schwartz, 1996), (e) book

¹³ For the first hour of trading, we construct volatility measures based on all available information at the time of the order (i.e., using time-windows less than an hour). Deleting the first hour of trading does not alter our results.

imbalance signals less competition on same side relative to opposite side of the book, and (f) the limit price of the previous order (a proxy for omitted market conditions) is less aggressive. The impact of market volatility and industry volatility is not significant.

We focus on the coefficient estimates of indicator variables, *DayMinus5* to *DayMinus1*. Note that the least aggressive order is categorized as “7” and the most aggressive order is categorized as “1”. Thus a negative (positive) *DayMinus3* coefficient supports the hypothesis that order flow observed on event Day [-3] is more (less) price aggressive than those observed on control days for the same firm. Since the research design attributes abnormal order flow on Day [-3] to informed agents, a negative *DayMinus3* coefficient suggests that informed agents use more aggressively priced orders to build the positions.

For buy orders submitted before positive events (column (1)), all the five coefficients corresponding to Days [-5,-1] are negative, and *DayMinus3* and *DayMinus1* coefficients have t-statistics below -2.0. Overall the cumulative effect coefficient is -0.56 (see Panel B, t-statistic=-2.71) implying that informed traders submit *more* aggressively priced buy orders before positive events. For sell orders submitted before negative events (column (2)), four of the five coefficients corresponding to Days [-5,-1] are positive and the *DayMinus5* coefficient is statistically significant (t-statistic=2.23). The cumulative effect coefficient of 0.24 (t-statistic=1.65) suggests that informed traders submit *less* aggressively priced sell orders before negative events. The “T-statistic of the difference between positive and negative events” tests the null hypothesis that the event effects for positive and negative news are the same. The highly significant t-statistic (= -3.16) suggests that informed traders strategies are influenced by the direction of the news. The asymmetry in informed trader strategies before positive and negative events is a novel contribution of the study.¹⁴

In terms of economic significance, the cumulative effect coefficient in linear specification is interpreted as the change in price aggressiveness on Days [-5,-1] relative to control days [-30,-10] after accounting for other determinants of price aggressiveness. Focusing on all positive events (column (1)),

¹⁴ The lack of a clear pattern of price aggressiveness between Day [-5] and Day [-1] points to the possibility that informed agents are not certain about the exact timing of the news release.

the cumulative coefficient -0.56 implies that, relative to the average price aggressiveness of 5.18 observed on control days, price aggressiveness *increases* on Days [-5,-1] by 10.8%. For negative events, a similar analysis indicates a *decrease* in price aggressiveness of 4.8%. We also examine the distribution of the seven categories of price aggressiveness on event days relative to control days. The observed changes in price aggressiveness point to a 8.12% increase in the frequency of aggressive orders (categories 1 to 4) on Days [-5,-1] for positive news versus a 4.67% decrease in the frequency of similar aggressive orders on Days [-5,-1] for negative news.

3.3. The speed of price discovery before positive and negative events

Microstructure models predict that stock prices incorporate more information when informed agents use aggressive orders before news releases. In this section, we investigate asymmetry in learning and price discovery before positive and negative events using “unbiasedness regressions” following Biais, Hillion, and Spatt (1999). For each half-hour time period (i) in interval Days [-5,-1], we regress the return from close on Day [-6] to close on Day [+1] on the return from close on Day [-6] to end of time period I , as follows:

$$ret_{[-6,+1]} = \alpha + \beta \cdot ret_{[-6,I]} + \varepsilon_i \quad (2)$$

We separately estimate the cross-sectional regression for each time period I , and calculate confidence intervals based on the standard error of the slope. Barclay and Hendershott (2003) interpret the slope of the unbiasedness regressions as a signal:noise ratio. According to the learning hypothesis, the slope will move closer to one as the pre-event security price reflects the post-event price with increasing precision.

Fig. 2 presents the slope and the confidence intervals for positive and negative events. At the beginning of interval [-5,-1] for both types of events, the slope is not significantly different from zero. In the case of positive events, towards the end of the interval [-5,-1], the slope is significantly different from zero, and not significantly different from one. Interpretations are similar from the bar chart that compares the root mean square error (RMSE) of the unbiasedness regression with variance of return from close on Day [-6] to close on Day [+1]. As the announcement day gets closer, the bar chart shows that the residual

variance declines. Thus, we cannot reject the *learning hypothesis* that informational efficiency of the pre-event stock price is no different from the post-event stock price before positive events.

In the case of negative events, the slope of the unbiasedness regression is significantly lower than one, and in fact, does not reach one by the close on Day [-1]. Similarly, the bar chart indicates that residual variance relative to return variance from close on Day [-6] to close on Day [+1] remains high. Thus, security prices incorporate less “news” before negative events than positive events.

In a related analysis that links informed trader strategies to the speed of price discovery, we estimate a cross-sectional regression of the event i 's CAR_ratio , defined as $CAR [0,+1]/CAR [-5,+1]$, on event i 's cumulative effect coefficient ($AGG_CumEffect$) from the event-by-event regression (Eq. (1)). A higher CAR_ratio indicates that stock prices incorporate less news before the event. Further, since the least aggressive order is categorized as “7” and the most aggressive order is categorized as “1”, a higher value of $AGG_CumEffect$ indicates that informed traders employ less price aggressive orders before the event. The regressions results are as follows:

$$CAR_ratio_i = \alpha + \beta AGG_CumEffect_i + \varepsilon_i \quad (3)$$

	0.557	0.449
<i>t-stat</i>	(3.17)	(2.13)

The highly significant, positive coefficient on $AGG_CumEffect$ suggests that stock prices incorporate more "news" when informed agents employ more price aggressive orders before news release.

3.4. Opportunity costs of non-execution of limit orders before positive and negative events

The price discovery analysis indicates that stock prices drift more in the direction of news before positive events than negative events. This result has important implications for the informed trader's choice of a market order versus a limit order. A drift in stock price in the direction of the signal lowers the informed traders' probability of execution of a limit order because prices tend to gravitate away from the limit price. To prevent limit orders from becoming stale, the informed trader has to keep offering better prices, implying it is more costly to employ limit orders before positive events than negative events.

In this analysis, we estimate the cost of non-execution of a limit order before positive and negative events using the implementation shortfall framework proposed by Perold (1988) and implemented by Harris and Hasbrouck (1996) and Griffiths, Smith, Turnbull, and White (2000). Prior studies note that each order is associated with two cost components: (a) the effective spread cost, which relates to portion of the order that is executed, is measured as the appropriately signed difference between the fill price and the quote mid-point at the time of order submission, and (b) the opportunity cost, which relates to the portion of the order that goes unfilled, is the appropriately signed difference between the closing price on order expiration or cancellation date and quote midpoint at the time of order submission. The implementation shortfall cost for an order is the weighted sum of effective spread and opportunity cost, where weights are the proportion of order size that is filled and unfilled, respectively.

Table 3 presents regression coefficients of implementation shortfall costs on the prevailing market conditions and order attributes. For a limit order that goes unfilled, the effective spread cost is zero. Thus the effective spread regression (columns (5) and (6)) includes only orders with either partial or full execution (fill rate > 0%). Consistent with Harris and Hasbrouck (1996), we find that orders that are more aggressively priced are associated with higher effective spread cost. However, after controlling for price aggressiveness, the incremental effective spread cost on Days [-5,-1] relative to control days, as reflected in the cumulative effect coefficient, is not statistically significant. Thus we do not find evidence that trading costs, conditional on execution, are significantly higher before unscheduled events.

We next focus on coefficients of opportunity cost regressions reported in column (7) and (8). For an order that is fully executed, the opportunity cost is zero; thus the analysis only considers orders with partial or full non-execution (fill rate < 100%). When a limit order goes unfilled, the informed trader might need to transact later at a less advantageous price. The opportunity cost, which captures the cost of the delayed execution, is positive if the stock price moves higher (lower) after a buy (sell) limit order is placed. The results in columns (7) and (8) suggest that hidden orders are associated with smaller opportunity costs, which is consistent with Bessembinder, Panayides, and Venkataraman's (2009) conclusion that hidden orders are primarily used by uninformed agents to control order exposure risk.

Results in Table 3 on opportunity costs suggest that employing limit orders on Days [-5,-1] before positive events is costly. For buy orders (column (7)), the cumulative effects coefficient is 0.32 (t-statistic=2.17), implying that informed traders face a higher opportunity cost when a limit order goes unfilled before a positive event relative to control days. The finding is consistent with the price discovery pattern that stock prices incorporate a portion of the good news before positive events (i.e., stock prices drift higher before a positive event). In contrast, for sell orders (column (8)), the opportunity cost is negative, implying that stock prices do not gravitate away from the limit price before negative events and that informed traders whose limit orders go unfilled need not transact later at a less advantageous price. The difference in opportunity costs between positive and negative events is statistically significant (t-statistic=2.79) and leads to a difference in implementation shortfall costs (t-statistic=2.62) reported in columns (3) and (4).

4. Modelling informed trader's strategy

The results thus far point to an asymmetry in order flow before positive and negative events. Prior to positive events, the proportion of aggressively priced orders is increasing, while before negative events the proportion is actually decreasing.¹⁵ Further, when a limit order goes unfilled, we estimate large opportunity costs prior to positive events but not negative events, indicating that employing an aggressive strategy before positive events is reasonable. The goal of this section is to provide a falsifiable economic rationale for these results. That is, to provide a model that (i) matches the findings, and (ii) provides novel empirical predictions that are outside our initial findings.

The natural candidate for an explanation of any form of asymmetry in trading patterns between positive and negative news is costly short selling. This is our starting point. In Kaniel and Liu (2006), the lower the probability that the next trader is informed, the more likely the current (informed) trade submits a limit order. We believe that an amalgam of the economics that underlies Kaniel and Liu (2006) together

¹⁵ A related literature on block trading examines buy-sell *price impact* asymmetry; that is, the well-known result that block purchases of securities convey more information than block sales of securities (see Kraus and Stoll (1972), Keim and Madhavan (1996)). Saar (2001) motivates the idea that informed sellers could trade for liquidity reasons while informed buyers do not, which is a source of price impact asymmetry.

with buy-sell asymmetry in competition that is introduced by costly short selling, as in Diamond and Verrecchia (1987), is a good foundation to build our framework.

We study a two period Glosten and Milgrom (1985) model with competitive market makers. In the first round of trading, the active (whether noise or informed) trader may submit a limit order.¹⁶ Informed traders have identical information and therefore always trade in the same direction; buy if the event is positive and sell if the event is negative. Hence, the probability of execution of informed limit order is inversely related to the mass of informed traders. Kaniel and Liu show that if the mass of informed traders is sufficiently small, then an informed trader in round one finds it optimal to submit limit orders.

In our model, the mass of informed trading is endogenous and non-trivial when the event is negative. Specifically, in the second round of trading, if the event is negative, constrained informed traders may prefer to abstain if the borrowing costs are high or the event is small. On the other hand, when the event is positive, all informed traders, constrained or not, are active. Thus, our model replicates the result in Kaniel and Liu (2006) in a non-symmetric manner: informed traders may use limit order when the event is negative, but avoid limit orders when the event is positive.¹⁷

4.1 The Model

We consider an *event period* that contains two trading rounds. The timing of the model is as follows: (i) Market makers post bid and ask prices for the first round of trading; (ii) A trader is picked at random from a large pool of potential traders, and submits an order (either a market or a limit order); (iii) When round two commences, market makers refresh their quotes; (iv) A second trader is then picked at random and submits a market order; (v) The asset liquidates.

We denote the asset value by $\tilde{v}$ and assume it is a 0-1 random variable.¹⁸ We let H_0 denote the history of trade up to the start of the event period, and let

¹⁶ Since the game ends after the second round, there is no point in submitting a limit order in the second round.

¹⁷ The four differences between our model and Kaniel and Liu's model are described in footnotes (18)-(21).

¹⁸ The two point distribution simplifies our analysis and is quite common in the literature. However, this assumption

$$p_0 \equiv P_0(\tilde{v} = 1) \equiv E_0 \tilde{v} \equiv E[\tilde{v}|H_0] = P(\tilde{v} = 1|H_0) \quad (4)$$

be the prior. Note that before the history of trades unfolds, p_0 itself is a random variable. However, during the event period, the prior, p_0 , is common knowledge and for the remaining of this section, we treat p_0 as a parameter.

A *positive event* is $\{\tilde{v} = 1\}$ and a *negative event* is $\{\tilde{v} = 0\}$. We say that an event is *small* when the magnitude of $\tilde{v} - p_0$ is small. That is, a positive event is small when p_0 is close to one, and a negative event is small when p_0 is close to zero. When the event is small, the potential gains of informed traders are small.

As in all probability selection models, we need to specify the types of potential traders and their respective masses. There are all together four groups of potential traders in our model. There are two groups of noise traders; market order traders and limit order traders. There are also two groups of informed traders; unconstrained and constrained informed traders. The distinction is according to whether or not the traders belong to the firm's investor base. Unconstrained investors already own the stock before the event period. Constrained investors do not own the stock, and can only sell when they pay the borrowing costs associated with short selling.

To simplify our analysis, we assume constrained informed traders are not present in the first round. This assumption can be interpreted as information leakage. Initially (i.e., first round of trading), information is acquired by traders who are interested in the stock. Later (i.e., second round of trading), the information leaks to traders that are not part of the firm's investor base; i.e., constrained traders. That said, this model is not about information leakage, and the assumption is made for simplification.¹⁹

is a deviation from Kaniel and Liu, who assume that the asset value is drawn from a continuous and symmetric distribution. In Kaniel and Liu, informed traders use limit orders when the asset value is sufficiently close to the unconditional expected value of the asset (i.e., in equilibrium, informed traders use limit orders when the value is between the bid and ask prices, or barely outside the quotes). Also in our model, informed traders will end up using limit orders if the value is sufficiently close to the expected value of the asset (though an artifact of the two point distribution assumption is that the asset value is always outside the quotes). Thus, the choice of support is benign.

¹⁹In Kaniel and Liu there is a positive probability of information leakage, though they assume that in such a case everyone learns the information.

It is natural to assume that the mass of each group of traders is related to the size of the firm's investor base. We let $\beta \in (0,1)$ be a measure of the size of the investor base. In each trading round, the mass of noise traders who submit market orders is $2\beta z$, each of these traders is equally likely to buy or sell. In the first round of trading, the mass of noise traders who submit limit orders is $2\beta l$, each of these traders is equally likely to submit a buy or a sell limit order. We assume that noise limit order traders are absent from the second round of trading.²⁰ The total mass of informed traders in the first round is $\beta\mu$, consistent with our assumption that only unconstrained informed traders are present in the first round. The total mass of informed traders in the second round depends on whether or not constrained informed traders participate. Specifically, we write the mass of informed traders in the second round as

$$\tilde{\mu} = I_{\{\tilde{v}=1\}}\mu + I_{\{\tilde{v}=0\}}\mu^- = \tilde{v}\mu + (1 - \tilde{v})\mu^- \quad (5)$$

where μ^- has to be determined in equilibrium. Specifically, if constrained informed traders choose to abstain, then $\mu^- = \beta\mu$, and if they short sell, then $\mu^- = \mu$.

Let a_i and b_i be the posted ask and bid prices in the i th round of trading, $i \in \{1,2\}$. We let O_i denote the order in round i ,

$$O_1 \in \{MB, MS, (LB; PB), (LS; PS)\}$$

$$O_2 \in \{MB, MS\}$$

where MB , MS , LB , LS represent a market buy, market sell, limit buy, and limit sell, respectively, and PB and PS are limit buy and limit sell prices, respectively.

Our notion of equilibrium requires that the following, (6)-(16), conditions hold. First are the rationality conditions for the market makers quotes in the first trading round.

$$a_1 = E_0[\tilde{v}|O_1 = MB] \quad (6)$$

$$b_1 = E_0[\tilde{v}|O_1 = MS] \quad (7)$$

Next, we require that limit prices satisfy

²⁰ Noise limit orders provide camouflage to the informed traders' limit orders. Limit orders placed in the second round never execute, so we choose to turn off the presence of noise limit order traders in the last round. Kaniel and Liu choose to assume that in the last round of trading, the mass of noise limit order traders is added to the mass of noise market order traders, which increases the proportion of noise trading in the last round. Qualitatively, the choice of assumption has no impact on the asymmetry between positive and negative events.

$$PS = E_0[\tilde{v}|O_1 = (LS; PS), O_2 = MB] \quad (8)$$

and

$$PB = E_0[\tilde{v}|O_1 = (LB; PB), O_2 = MS] \quad (9)$$

To see why these conditions are natural, consider (8), which corresponds to the offer (ask) side. If the trader posts at a price strictly higher than the conditional expectation, then market makers will undercut and the limit order will never execute. On the other hand, if the trader posts at a price strictly lower than the conditional expectation, it would be a needless concession. Therefore, condition (8) is natural. A similar argument justifies condition (9).²¹

We also require that, in round one, an informed trader optimally chooses between a market order and a limit order, taking into account the execution risk of the limit order. That is, in a positive event, an informed trader prefers a buy market (resp. limit) order over a buy limit (resp. market) order at PB only if

$$(1 - a_1) \geq (1 - PB) \frac{\beta z}{2\beta z + \mu} \quad (\text{resp. } \leq) \quad (10)$$

and, in a negative event, an informed trader prefers a sell market (resp. limit) order over a sell limit (resp. market) order at PS only if

$$b_1 \geq PS \frac{\beta z}{2\beta z + \mu} \quad (\text{resp. } \leq) \quad (11)$$

The fraction that appears at the right hand sides of (10) (resp. (11)) is the execution probabilities of a buy (resp. sell) limit order, conditional on positive (resp. negative) event.

Lemma 1. *Consider the first round of trading. If informed traders use limit orders, then (i) informed traders and noise traders use the same limit prices, and (ii) informed traders are indifferent between limit orders and market orders.*

²¹ The limit prices that satisfy (8) and (9) are the ones that minimize the expected losses from trade of the group of noise limit order traders, subject to the constraint that the limit order is the best quote in the second round. Kaniel and Liu also posit the same price, though they also verify a *participation constraint*: noise limit order trader should not prefer market orders. In their Lemma 6, Kaniel and Liu show the constraint is never binding. Also here the participation constraint is never binding, and we choose not to include it as part of our definition of equilibrium.

The proof, found in Appendix B, relies only on mutual consistency between conditions (6)-(11). Thus, the lemma is a partial equilibrium result. The first part of the lemma implies that to find the equilibrium, we need to find a single pair of deterministic limit prices; PS and PB . Moreover, because both noise and informed traders use the same non-random limit prices, limit prices do not contain information. Thus, we don't lose any generality if we assume that the first round order, O_1 , is in the set $\{MB, MS, LB, LS\}$, and rewrite (8) and (9)

$$PS = E_0[\tilde{v}|O_1 = LS, O_2 = MB] \quad (8')$$

$$PB = E_0[\tilde{v}|O_1 = LB, O_2 = MS] \quad (9')$$

The second part of the lemma says that limit orders cannot be strictly optimal for informed traders. This means that whenever informed traders use limit orders, they may as well use market orders. Let m_p and m_n be the probabilities that an informed trader submits market orders when the event is positive and negative, respectively. The probabilities that an informed trader submits limit orders for positive and negative events are $1 - m_p$ and $1 - m_n$, respectively. Since limit orders cannot be strictly optimal, Lemma 1 implies that we don't lose any generality if we replace (10) and (11) with

$$(1 - a_1) \geq (1 - PB) \frac{\beta z}{2\beta z + \mu} \quad \text{with } > \text{ only if } m_p = 1 \quad (10')$$

$$b_1 \geq PS \frac{\beta z}{2\beta z + \mu} \quad \text{with } > \text{ only if } m_n = 1 \quad (11')$$

We turn now our attention to the second round of trading. The quotes in the second round should depend on the type of order that was submitted in the first round, and we therefore denote them by $a_2(O_1)$ and $b_2(O_1)$. We require that

$$a_2(O_1) = E_0[\tilde{v}|\tilde{O}_1 = O_1, O_2 = MB] \quad (12)$$

$$b_2(O_1) = E_0[\tilde{v}|\tilde{O}_1 = O_1, O_2 = MS] \quad (13)$$

Note that the above is consistent with conditions (8')-(9'); namely $a_2(LS) = PS$, and $b_2(LB) = PB$. The subtle point is that if $O_1 = LS$, then $a_2(LS)$ is a limit-order trader's quote while $b_2(LS)$ is the market makers' quote. Similarly, if $O_1 = LB$, then $b_2(LB)$ is a limit-order trader's quote, while $a_2(LB)$ is

the market makers' quote. If O_1 is a market order, then both quotes are the market makers' quotes. Conditions (12)-(13) state that regardless who is quoting, the quotes satisfy the standard rationality condition in round two.

Finally, during a negative event, constrained investors are present only if the bid is greater than the borrowing costs. To emphasize the dependency of the mass of informed trading during negative event on the posted bid, we write $\mu^-(b_2)$. We denote the borrowing costs by c , and require that

$$b_2 > c \Rightarrow \mu^-(b_2) = \mu \quad (14)$$

$$b_2 < c \Rightarrow \mu^-(b_2) = \beta\mu \quad (15)$$

$$b_2 = c \Rightarrow \mu^-(b_2) \in [\beta\mu, \mu] \quad (16)$$

Condition (16) is relevant only in a knife edge situation, in which the constrained investors are indifferent. In that case, any value for $\mu^-(b_2)$ in the interval $[\beta\mu, \mu]$ is acceptable.

Hereafter, an equilibrium is a tuple

$$(a_1, b_1, m_p, m_n, a_2(\cdot), b_2(\cdot), PB, PS, \mu^-(\cdot))$$

such that for every $O_1 \in \{MB, MS, LB, LS\}$, the elements of the tuple

$$(a_1, b_1, m_p, m_n, a_2(O_1), b_2(O_1), PB, PS, \mu^-(b_2(O_1)))$$

satisfy conditions (6)-(16).

Lemma 2 below shows that our model is not vacuous: If borrowing costs are high, the investor base is narrow, and the event is small and negative, then informed traders may submit sell limit orders (i.e. $m_n < 1$), while constrained informed traders abstain.

Lemma 2. *Assume the event is negative.*

1. *If borrowing cost is large, or event is small, such that $p_0 < c$, then constrained informed traders abstain in the second round (i.e., $\mu^-(b_2(LS)) = \beta\mu$).*
2. *If (i) event is small such that $p_0 < \mu/(z + 2\mu)$, (ii) constrained informed traders abstain (i.e., $\mu^-(b_2(LS)) = \beta\mu$), and (iii) investor base is narrow (i.e., β is sufficiently small), then informed traders in the first round use limit orders (i.e., $m_n < 1$).*

The proof of Lemma 2 is in Appendix B.

4.2. Empirical predictions

Recall that $p_0 = E[\tilde{v}|H_0]$, the prior, depends on the realization of the history up to the start of the event window. Our empirical prediction will be based on the assumption that the prior is a symmetric random variable, $\tilde{p}_0$. That is, positive events are as likely as negative events, and small positive events are as likely as small negative events. In particular, $E\tilde{v} = 1/2$. Thus, the only source of asymmetry between positive and negative events in our model is the need of constrained informed traders to borrow the shares during negative events.

To emphasize the dependency of the equilibrium on the realization of $\tilde{p}_0$, we write the equilibrium as

$$(a_1(p_0), b_1(p_0), m_p(p_0), m_n(p_0), a_2(p_0, O_1), b_2(p_0, O_1), PB(p_0), PS(p_0), \mu^-(b_2(p_0, O_1)))$$

for every $O_1 \in \{MB, MS, LB, LS\}$.

In preparation for our Theorem 1, in Lemma 3 we compare the probability of informed sell market orders during negative events and informed buy market orders during positive events. To control for the size of the event, the comparison is carried at symmetric values of the prior. That is, we compare $m_n(p_0)$ with $m_p(1 - p_0)$.

Lemma 3. *Consider the equilibria that correspond to $\tilde{p}_0 = p_0$ and to $\tilde{p}_0 = 1 - p_0$. Then,*

1. $m_n(p_0) \leq m_p(1 - p_0)$.
2. *If both (i) $\mu^-(b_2(p_0, LS)) < \mu$ and (ii) $m_n(p_0) < 1$ hold, then $m_n(p_0) < m_p(1 - p_0)$; i.e. the inequality is strict.*

Theorem 1. *Assume that $\tilde{p}_0$ is drawn from a symmetric distribution*

1. *Unconditionally (i.e. without reference to the realization of $\tilde{p}_0$), the probability of observing a sell limit order when the event is negative is as high as the probability of observing a buy limit order when the event is positive.*

2. [Asymmetry] *Consider small events (i.e. $\tilde{v} - \tilde{p}_0$ is close to zero). If borrowing costs are sufficiently high and the investor base is sufficiently small, then the probability of observing a sell limit order in a negative event is strictly greater than the probability of observing a buy limit order in a positive event.*
3. [Symmetry] *When the investor base is large, events are large, or borrowing costs are small, then the probability of observing a sell limit order in a negative event is similar to the probability of observing a buy limit order in a positive event.*

Both the proof of Lemma 3 and Theorem 1 are in Appendix B.

Our model predicts that short sale constraints lead to asymmetry in informed traders' strategies before positive and negative events. The empirical literature (see Harris and Hasbrouck, 1996; Bloomberg, O'Hara, and Saar, 2005; Hopman, 2007; Bershova, Stephens, and Waelbroeck, 2014) shows that the price impact of a market order is dramatically larger than the price impact of a limit order. This evidence is consistent with the patterns in Fig. 2 where stock prices incorporate more news before positive events than negative events. That is, the usage of more aggressive informed traders' strategies before positive events leads market participants to incorporate more news before positive events than negative events. We posit the following prediction that relates short sale constraints to price efficiency asymmetry before events:

4. [Price efficiency asymmetry] *The stock price incorporates more “news” before positive events than negative events, particularly when event is small, investor base is narrow, and borrowing costs are high.*

5. Cross-sectional evidence on order submission strategies, price discovery, and opportunity cost

The empirical literature on short sale constraints has identified a number of firm characteristics that are associated with the ease of short selling. To test model predictions, we examine informed traders' strategies and price discovery based on three cross-sectional measures of short sale constraints.

The first measure is the stock's membership in an index, which influences the broadness of the

investor base. Specifically, index firms tend to be larger, actively traded, and attract more attention from analysts and buy-side institutions (Nagel, 2005). Index stocks are also easier to borrow because index funds are active lenders of securities. We classify stocks based on membership in the SBF120 index.²²

The second measure is based on the availability of exchange-listed options contracts in a stock. When stocks are short constrained, Battalio and Schultz (2011) show that informed traders build short positions using option contracts. Hu (2014) shows that options market makers absorb the imbalance and hedge inventory risk by short selling the underlying stock. Since options market makers enjoy special exemptions on inventory-hedging stock trades, they are able to implement short positions at low cost, and via this mechanism, exchange-listed options lower short sale constraints. In our setting, one implicit assumption is that options market makers use aggressively priced orders to hedge inventories in the stock market. Data on exchange-listed options for sample firms are obtained from the SBF database.

The third measure is based on a unique institutional feature on Euronext Paris that affects the ease of short selling. For certain *eligible* stocks, the Deferred Settlement Service (“Service de Règlement Différé”, henceforth SRD) allows investors the convenience to take long and short positions with deferred settlement of the trade until the end of the month (see Foucault, Sraer, and Thesmar, 2011). Stocks that are eligible for SRD-facility are selected by the exchange. An investor who wishes to sell short an SRD-eligible stock must flag the order as *deferred execution*. On executing the short sale, the broker effectively acts as a lender of the stock until the end of the month. In our setting, the SRD facility offers informed sellers a convenient way to short sell an SRD-eligible stock for an additional fee as long as they cover the short position by the end of the month.²³ Foucault, Sraer, and Thesmar (2011) report that short selling a stock that is not eligible for Euronext’s SRD facility is cumbersome because short sellers need to locate shares they want to sell in advance of executing a short sale. Information on SRD-eligibility for sample stocks in 2003 was made available by Euronext-Paris.

²² The SBF120 index represents a broader cross-section of stocks than the CAC40 Index (see Bessembinder and Venkataraman (2004)).

²³ For a typical retail order in June 2001, Foucault, Sraer, and Thesmar (2011) report that online brokers charge an additional fee of 0.20% of the order (with a minimum amount of 6.2 euros) for orders with deferred execution. In comparison, the median announcement return for negative events reported in Table 1 is -2.57%.

Our model predicts that informed traders' strategy depends on the magnitude of the event. Informed sellers have incentives to locate difficult-to-borrow shares when negative news is large. When the negative news is small, the benefits from short selling might not out-weigh the borrowing cost. For this reason, informed sellers who do not already own the stock might abstain from trading. We classify events as large when the absolute value of announcement day return exceeds 5%.²⁴

Table 4 presents the main results on the cross-sectional analysis of events. Panel A reports the results for index membership and SRD-eligibility and Panel B reports the results for options listing and announcement return. The model predicts that informed traders employ more aggressive strategies for all positive events (columns (1), (3), (5) and (7) in Panels A and B), negative events without short constraints (column (2) in Panels A & B, and column (6) in Panel A), and large negative events (column (6) in Panel B). Additionally, the model predicts that informed traders employ less price aggressive strategies before negative events in short constrained stocks (column (4) in Panels A & B, and column (8) in Panel A) and small negative events (column (8) in Panel B). For ease of exposition, we refer to the former category as *market order (MO)* equilibrium events and the latter category as *limit order (LO)* equilibrium events.

5.1. The cross-section of informed traders' strategy on price aggressiveness

In Panels A.1. and B.1. of Table 4, we present univariate statistics on daily market:limit ratio on event and control days. For all limit order (LO) events, we observe a decrease in the market:limit ratio. As an example, in column (4) of Panel A, the percentage of market:limit orders declines from 51.5% on control days to 47.4% on events days, indicating that informed traders employ a higher percentage of limit orders relative to market orders to build positions before the event. Similar patterns are observed for other limit order (LO) events reported in column (8) of Panel A, and columns (4) and (8) of Panel B. In contrast, for all market order (MO) events, the market:limit ratio increases from control days to the event

²⁴ We also classify events as large (above median) and small (below median) after dividing the absolute value of announcement return by the stock's average bid ask spread on control days. With this measure, we attempt to incorporate the trade-off between using market orders and paying the spread versus using limit orders and receiving the spread. The sign (and statistical significance) of regression coefficients are similar to those reported in the paper.

days indicating that informed traders employ a higher percentage of market orders relative to limit orders. These results provide preliminary support for the cross-sectional predictions of the model.

In Panels A.2. and B.2. of Table 4, we report the number of trades on control days and the percentage change (and the associated test of difference) in number of trades on event days relative to control days. We find that the limit-order (LO) events are associated with the smallest increase in trading activity on event days relative to control days. As an example, based on index membership, the increase in trades for negative events in non-index firms is 11.9% (column (4) of Panel A) while the increase in trades in columns (1) to (3) of Panel A exceeds 20%. The smaller increase in trading activity observed before negative events for short constrained stocks, and for negative events with small news release, is consistent with the model's assumption that constrained informed sellers abstain when borrowing costs do not exceed the benefits of short selling.

In Panels A.3. and B.3. of Table 4, we report cumulative effects coefficient based on an event-by-event price aggressiveness regression and then aggregated using the Bayesian framework described in Appendix A. The results provide strong support for the model predictions. For all market order (MO) events, the cumulative effects coefficient is negative, which indicates that informed order flow is associated with more price aggressive orders. In nine out of 12 columns, the coefficients are statistical significant at the 5% level, and further, in two columns, the coefficients are significant at the 10% level.

For all limit order (LO) events, we observe a positive cumulative effects coefficient, which indicates that informed order flow is associated with less price aggressive orders. In two of four columns, the coefficients are statistical significant at the 5% level, while for the remaining two columns, the coefficients are significant at the 10% level.

Additionally, in the case of short sale constrained firms, we find support for an buy-sell *asymmetry* in order submission before positive and negative events. The difference in cumulative effect coefficient between positive and negative events is highly significant for non-index stocks, non SRD-eligible stocks, stocks without listed options, and events with small announcement return. Further, the results support *symmetry* in order submission between positive and negative events when informed agents

anticipate competition. Specifically, the difference in cumulative effects coefficient between positive and negative events is not statistically significant for index stocks, SRD-eligible stocks, stocks with listed options, and events with large announcement returns. Thus we observe a buy-sell asymmetry for stocks that are difficult to short but not for stocks that are easier to short. These results provide strong empirical support for the novel testable predictions of the model.

The economic significance of these results is large. As an example, for large positive events, we estimate an increase of 27.78% in buy order aggressiveness on event days relative to control days, which reflects a 43.59% increase in the frequency of aggressive buy orders (categories 1-4). The corresponding statistics are 6.37% and 4.67%, respectively, for small positive events. For small negative events, the positive cumulative coefficient translates into a decrease of 4.78% in sell order price aggressiveness and a 4.78% decrease in frequency of aggressive sell orders.

We also examine trading activity in the options market for stocks with listed options. While we observe no significant change in daily options trades relative to control days prior to positive events, we observe an increase in daily options trades prior to negative events. In particular, options activity is nearly twice as large and statistically different on Day [-1]. The higher activity in options market prior to negative events suggests that some informed sellers use options to build short positions, consistent with Battalio and Schultz (2011).

Our results are consistent with theoretical papers that model a dynamic limit order book with strategic traders. For example, Goettler, Parlour, and Rajan (2009) predict that informed traders use limit orders when volatility is low and Rosu (2014) predicts that informed traders use limit orders when realized signals are close to their expected value. In support of these predictions, and consistent with our model, we find that informed traders use limit orders before negative events with small announcement effects. However, consistent with our model, and inconsistent with Goettler, Parlour, and Rajan (2009) and Rosu (2014), we show that informed traders use market orders before positive events with small announcement effects. That said, when event is small, the orders that the informed traders submit are less aggressive than the orders when event is large.

Stocks with small investor base and high borrowing costs can be less liquid. Informed traders face a smaller explicit cost of using market orders (bid-ask spread) in liquid stocks; however the higher price impact of market orders as compared to limit orders can increase the total cost of building a position. To investigate the interplay between short sale constraints and stock liquidity, we separately examine asymmetry in informed trader behavior for liquid and illiquid stocks. We classify a stock to be liquid if it is placed in the lowest tercile based on percentage bid-ask spreads in the control period. In unreported results, the statistical tests based on the cumulative effects coefficient from event-by-event regressions support asymmetry between limit order (LO) and market order (MO) events based on short sale constraints for both liquid and less-liquid stocks.

Table 4 reports the number of events for each subsample. Although many index stocks have exchange-listed options and are eligible for the SRD facility, we find that the overlap leaves sufficient room for independence across these cuts of the data. For example, focusing on index stocks, 10 among 24 positive event stocks and five among 16 negative event stocks do not trade with listed options. Similarly, focusing on stocks without listed options, 13 among the 44 positive events and nine among 28 negative events are eligible for SRD facility.

5.2. *Short sale constraints and the speed of price discovery*

We further investigate the impact of short sale constraints on the extent of learning and price discovery before corporate events. We estimate separate cross-sectional unbiasedness regressions for different subsamples, and calculate confidence intervals based on the standard error of the slope. Fig. 3 reports the slope and the confidence intervals for four subsamples based on SRD-eligibility: (a) positive events for SRD-eligible firms, (b) negative events for SRD-eligible firms, (c) positive events for SRD-ineligible firms, and (d) negative events for SRD-ineligible firms. Similar to Biais, Hillion and Spatt (1999), we also plot the ratio of the root mean square error (RMSE) of each unbiasedness regression to the unconditional standard deviation of the close-to-close return (ret_{cc}).

For the market order events (i.e., (a), (b) and (c) above), the slope of the unbiasedness regression by the end of the pre-event interval is not significantly different from one. We cannot reject the learning hypothesis that informational efficiency of the pre-event stock price is no different from post-event stock price. In contrast, for limit order events (i.e., (d) above), the slope is significantly less than one, and in fact, not significantly different from zero. The patterns based on index membership and the availability of stock options are similar but are not reported in the interest of brevity. Overall, the patterns suggest that limit order events incorporate less news before announcement than market order events.

Results based on the regression residuals also point to lower price discovery for limit order events than market order events. For market order events, the ratio drops substantially at the end of the interval [-5,-1] indicating that informed traders' private information is incorporated in prices before announcement. For limit order events, the ratio does not drop substantial at the end of the interval [-5,-1]. We are unable to reject the hypothesis that noise trading dominates the trading process.

5.3. Short sale constraints and opportunity cost of non-execution of limit orders

The patterns in Fig. 3 suggest that stock prices incorporate more news before market order events than limit order events. These patterns raise the possibility that using limit orders before market order events is costly because if the limit order goes unfilled, the informed trader will transact later at a less advantageous price. To empirically estimate the cost of an unfilled limit order, we report cumulative effects coefficients of the opportunity cost regressions for sub-sample of events in Panels A.4. and B.4. of Table 4.

For all market order (MO) events (with the exception of column (2) in Panel B.4.), we observe a positive cumulative effects coefficient, which indicates that stock prices move higher before positive MO events (columns (1), (3), (5) and (7) of Panels A and B) and move lower before negative MO events (columns (2) and (6) of Panel A and column (6) of Panel B). In seven out of 12 columns, the MO coefficients are statistical significant at the 5% level, and further, in one column, the coefficient is significant at the 10% level. These results suggest that stock prices tend to gravitate away from the limit

price and that limit orders face a higher risk of becoming stale. For all limit order (LO) events, we observe a negative cumulative effects coefficient, although only one of four coefficients is statistically significant. The results do not support that stock prices gravitate away from limit price, suggesting that using limit orders to build positions before *LO* events is a reasonable strategy. Additionally, the difference in cumulative effects coefficients between positive and negative events is (a) highly statistically significant for non-index stocks, non SRD-eligible stocks, stocks without listed options, and small event returns, and (b) not statistically significant for index stocks, SRD-eligible stocks, stocks with listed options, and large events.

Overall, the results in Section 5 support the theoretical prediction that informed agents' strategies affect the process of price formation in financial markets. Informed agents employ aggressive orders (i.e., price aggressiveness measure is negative) when they anticipate competition. Aggressive orders are easier to detect by market participants, who move stock prices in the direction of signal (i.e., opportunity cost of non-execution is large) before public announcement. When faced with less competition, informed agents employ passive strategies (i.e., price aggressiveness measure is positive). Market participants have more difficulty in detecting passive strategies and therefore the stock price adjustment before public announcement is small (i.e., opportunity cost of non-execution is small). Our study provides empirical support for the mechanism that links informed trading strategies to the price discovery process.

6. Robustness analysis

The analysis thus far classifies events as positive and negative news based on announcement period (Days [0,+1]) cumulative abnormal returns (CAR). We find that results are similar if events are classified as positive and negative based on longer period (Days [-5,+1]) CAR. We also classify events based on abnormal imbalance observed before events between buy and sell order flow (non-marketable as well as marketable), as follows: $[\text{Buy Volume} - \text{Sell Volume}]_{\text{EVENTDAYS}} - [\text{Buy Volume} - \text{Sell Volume}]_{\text{CONTROLDAYS}}$. Specifically, we aggregate the displayed order size each day across all buy orders and all sell orders and then calculate the average daily difference during event and control periods,

respectively. Events are classified as positive when abnormal imbalance $>$ zero and negative when abnormal imbalance $<$ zero. We observe a significant overlap between the classifications based on announcement period CAR and order imbalance (87 of the 101 events), and near perfect overlap based on longer period CAR and order imbalance (97 of the 101 events). Moreover, for *limit order* events classified using announcement period CAR, the overlap with other classifications is near perfect.²⁵ Thus the important result that informed agents use limit orders when they anticipate less competition is not sensitive to the classification scheme.

In Table 5, we present robustness analyses based on several alternative specifications of the price aggressiveness regression. In Panels A.1 and B.1, the event's impact is captured by a single event day coefficient, which equals one for orders submitted over the interval $[-5,-1]$, and equals zero otherwise. In comparison to estimating separate event day coefficients, a single coefficient imposes the constraint that informed trading intensity is the same on each of the five days before the news release. We estimate the model for each separate event and aggregate coefficients across events using the Bayesian framework described in Appendix A.

When sample events are not clustered in calendar time, it is common in the event study literature to estimate the model on an event-by-event basis (see Binder, 1985; Thompson, 1985, 1995). This is because the efficiency gain from a panel specification, which imposes additional constraints on parameter estimates, is expected to be small since forecast errors across events are expected to be uncorrelated. Since our sample period of one-year is short, we present robustness analyses based on panel regression specifications in Table 5 (Panels A.2, A.3, B.2, and B.3). The panel specifications are based on individual event day coefficients in the interval $[-5,-1]$. Reported are cumulative effect coefficients and the t -statistics based on robust standard errors with event fixed effects and clustered standard errors by date. In

²⁵ For *limit order* events, the overlap in classification based on announcement period CAR and the order imbalance is as follows: 18 of the 19 events for SRD-ineligible stocks, 26 of the 27 events for non-index stocks, and 26 of the 28 events for stocks without listed options. The results using alternative classification schemes are available from the authors.

unreported analysis, we find that results are similar in a panel specification using a single event day coefficient for the interval [-5,-1].

In Panels A.2 and B.2, we report cumulative effects coefficient estimated separately for a pooled sample of events that are *similar* in terms of the direction of news and each of the three measures of short sale constraints. Pooling similar events helps incorporate the theoretical prediction that “true” event effects vary by firm attributes, such as short sale constraints, and event attributes, such as the magnitude of the announcement. It also helps alleviate to some extent the difficulty in panel analysis to disentangle the effect of true variation in parameter estimates across events from the variation that is attributable to estimation error. In Panels A.3 and B.3, we estimate the panel regression using the full sample of events and restrict the event day coefficients to be the same for *similar* events.

Results in Table 5 suggest that the theoretical predictions are strongly supported in the alternative model specifications. Specifically, for all *market order* events (columns (1)-(3) and (5)-(7) in Panels A and B), we obtain a negative cumulative effect coefficient, which indicates that informed order flow is associated with more price aggressive orders. We find that 28 out of 36 coefficient estimates are statistical significant at the 5% level, and further, six estimates are significant at the 10% level. For all *limit order* events, the cumulative effects coefficient is positive, which indicates that informed order flow is associated with less price aggressive orders. Six of the 12 coefficients are statistical significant at the 5% level, and three more coefficients are significant at the 10% level. Additionally, for all subsamples of short constrained firms and for events with small announcement effects, we find support for buy-sell asymmetry in trading strategy before positive and negative events.

We find that the alternative model specifications of the opportunity cost regressions yield results that are similar to those reported in Table 4. In the interest of brevity, these results are not reported in Table 5 but are available from the authors on request.

7. Conclusions

This study extends our understanding of how competition among informed agents influences the

choice of aggressive versus passive strategies to build positions before a news release. We further show that strategies of informed traders affect the process of price formation and the informational efficiency of prices before corporate events.

The intuition for the model is as follows. The presence of informed agents increases the execution risk on informed limit orders, which leads an informed agent to preempt and use market orders. Absent competition, an informed agent may prefer a limit order to a market order to lower the cost of building a position. We extend the Kaniel and Liu (2006) framework by incorporating short sale constraints that leads to an asymmetry in competition among informed agents before positive and negative events. When short selling is costly and the magnitude of negative news is small, our model predicts that informed agents use limit orders. When news is positive, informed agents always expect high execution risk for their limit orders and respond by using market orders.

For a sample of Euronext-Paris stocks, we find that the order mix of market and limit orders is influenced by anticipated competition among informed agents. Specifically, we present evidence of an asymmetry in price aggressiveness of informed order flow before positive and negative events. When short selling is costly, when the magnitude of news is small, or when investor base is not broad, informed sellers anticipate less competition and therefore use less price aggressive orders before negative events. In all other scenarios, informed traders anticipate competition and therefore use aggressive orders.

We further show that informed traders' strategies affect the informational efficiency of prices. In scenarios when informed agents use aggressive orders, we are unable to reject the learning hypothesis that informational efficiency of the pre-event security price is no different from the post-event security price. In contrast, when informed agents use less aggressive orders, we are able to reject that stock prices incorporate the negative news before announcement date. The key finding is that stock prices incorporate less news before negative events than before positive events, particularly in short constrained stocks.

Although our study focuses on informed trading in a centralized limit order market, our theory has implications for informed trading in fragmented markets, where the strategy space includes the choice of trading platforms with different levels of transparency. Some trading platforms such as Dark Pools are

designed to offer lower cost but with lower fill rates (see Zhu, 2014; Buti, Rindi, and Werner, 2014). Specifically, in the case of Dark Pools functioning as a crossing network, the probability of execution decreases as the number of traders participating on the same side of the market increases. In our theoretical model, this scenario is captured by anticipated competition among informed traders. As some informed traders recognize the low rate of execution in the crossing network, they route their orders to the transparent exchange, despite the higher cost of execution. The price impact of their orders in the transparent exchange increases the execution cost for informed traders remaining in the crossing network since the Dark Pool crossing price is based on quotations from the transparent exchange. Thus informed traders in fragmented markets face a risk-return tradeoff affected by competition that is similar to the one that we have modeled in a centralized market.

Our study points to a specific mechanism by which restrictions on short selling impede the flow of private negative information into prices (see Beber and Pagano, 2013). A short sale ban lowers competition among informed sellers and leads to a limit-order equilibrium which makes it difficult to detect informed order flow. This leads to lower price discovery before negative news relative to positive news. Our study is relevant for regulators interested in the efficacy of short sale regulation, particularly in markets around the world that continue to restrict or impede short sales in one form or another.

Appendix A: Aggregation method

Our aggregation method relies on a Bayesian framework attributable to DuMouchel (1994) and implemented by Bessembinder, Panayides, and Venkataraman (2009). The method assumes that, for each estimated firm i coefficient, β_i :

$$\hat{\beta}_i | \beta_i \sim i.i.d.N(\beta_i, s_i^2) \quad (A1)$$

$$\beta_i \sim i.i.d.N(\beta, \sigma^2) \quad (A2)$$

where N is the Gaussian distribution. The Newey-West corrected standard errors, s_i , are estimated using Generalized Method of Moments (GMM) with a Bartlett kernel and a maximum lag length of 10, and σ^2 is estimated by maximum likelihood.

Harris and Piwowar (2006) emphasize the desirability of assigning larger weights in the cross-sectional aggregation to those securities whose parameters are estimated more precisely. Consistent with the study's recommendation, the aggregated β is obtained from N individual firm estimates as -

$$\hat{\beta} = \frac{\sum_{i=1}^N \frac{\hat{\beta}_i}{(s_i^2 + \hat{\sigma}_{m.l.e}^2)}}{\sum_{i=1}^N \frac{1}{(s_i^2 + \hat{\sigma}_{m.l.e}^2)}} \quad (A3)$$

Assuming independence across firms, the variance of the aggregate estimate is:

$$Var(\hat{\beta}) = \frac{1}{\sum_{i=1}^N \frac{1}{(s_i^2 + \hat{\sigma}_{m.l.e}^2)}} \quad (A4)$$

where $\hat{\sigma}_{m.l.e}^2$ is the maximum likelihood estimator of σ^2 . The aggregate t-statistic is based on the aggregated coefficient estimate relative to the standard error of the aggregate estimate. This method

allows for variation across stocks in the true β_i and also for cross-sectional differences in the precision with which β_i is estimated.²⁶

²⁶ Note that the aggregated β estimate deals with the error-in-variable issue. It also corrects for the variability in the sample selection through $\hat{\sigma}_{m.l.e}^2$. The method does not control for dependence of estimation errors across events. We believe that this dependence should be small since the events are not clustered in time.

Appendix B: Proofs of Lemmas 1-3 and Theorem 1

Proof of Lemma 1. Consider the first part of the lemma. Assume by means of contradiction that informed traders post limit orders and their limit prices are different than the limit prices of noise limit order traders. Then the only price that satisfies conditions (8)-(9) is $PS = \tilde{v}$. But then, optimality conditions (10)-(11) imply that market orders are preferable to limit orders; a contradiction to the assumption that informed traders use limit orders.

Consider now the second part of the lemma. Assume that informed traders strictly prefer buy limit orders over buy market orders. That is, the right hand side of (10) is strictly greater than its left hand side. The right hand side is strictly smaller than $1/2$. If only noise traders submit buy market orders in round one, the left hand side of (10) is $a_1 = 1/2$, and we get a contradiction. In a similar manner we show that informed traders cannot strictly prefer sell limit orders over sell market orders.

Proof of Lemma 2. Consider the first part. From the equilibrium condition (15) it follows that we only need to show that

$$b_2(LS) < c$$

From equilibrium condition (8), we know

$$b_2(LS) = P_0(\tilde{v} = 1 | \tilde{O}_1 = LS, O_2 = MS) < E_0 \tilde{v} = p_0$$

where the inequality is because the order flow contains information. Thus, $p_0 < c$ implies $\mu^-(b_2(LS)) = \beta\mu$.

We turn our attention to the second part of the lemma. By way of contradiction, assume that in equilibrium $m_n = 1$. So the left hand side of the equilibrium condition (11') is

$$\begin{aligned} b_1 = E_0[\tilde{v} = 1 | \tilde{O}_1 = MS] &= \frac{\frac{\beta z}{2\beta z + \beta\mu + 2\beta l} p_0}{\frac{\beta z}{2\beta z + \beta\mu + 2\beta l} p_0 + \frac{\beta z + m_n \beta \mu}{2\beta z + \beta\mu + 2\beta l} (1 - p_0)} \\ &= \frac{zp_0}{zp_0 + (z + \mu)(1 - p_0)} \end{aligned}$$

On the other hand, the right hand side of the same equilibrium condition, (11'), is

$$\begin{aligned}
PS \frac{\beta z}{2\beta z + \mu^-(b_2(LS))} &= E_0[\tilde{v}|O_1 = LS, O_2 = MB] \\
&= E_0[\tilde{v}|O_2 = MB] \\
&= \frac{\frac{\beta z + \mu}{2\beta z + \mu} p_0}{\frac{\beta z + \mu}{2\beta z + \mu} p_0 + \frac{\beta z}{2\beta z + \mu^-(b_2(LS))} (1 - p_0)} \times \frac{\beta z}{2\beta z + \mu^-(b_2(LS))} \\
&= \frac{(\beta z + \mu) z p_0}{\beta z(\mu p_0 + 2z) + \mu(\mu p_0 + (1 + p_0)z)} \\
&\xrightarrow{\beta \rightarrow 0} \frac{z p_0}{\mu p_0 + (1 + p_0)z} \\
&> b_1 \quad \text{for } p_0 < \frac{\mu}{z + 2\mu}
\end{aligned}$$

The second equality holds because under the assumption that $m_n = 1$, a limit sell order in round one can come only from a noise limit order trader. In particular, observing a limit sell in round one is uninformative.

Thus, we see that for β sufficiently close to zero, equilibrium condition (11') is violated, thus we get a contradiction.

Proof of Lemma 3. Given an arbitrary choice of scalars m_p , m_n , μ^- , and b , the equilibrium conditions (6), (7), (8'), and (9') dictate what values a_1 , b_1 , PS , PB , and $b_2(LS)$ are consistent with the arbitrary choice of m_p , m_n and μ^- .²⁷ It is convenient to express these values in the following manner.

$$(1 - a_1) = f(1 - p_0, m_p) \tag{B1}$$

$$b_1 = f(p_0, m_n) \tag{B2}$$

²⁷In equilibrium, μ^- depends on the realized order at period one. Thus, in equilibrium, (B3) holds with $\mu^- = \mu^-(b_2(LB))$, and (A4) holds with $\mu^- = \mu^-(b_2(LS))$.

$$(1 - PB) \frac{\beta z}{2\beta z + \mu} = h(g(1 - p_0, m_p), \mu^-, \mu) \quad (B3)$$

$$PS \frac{\beta z}{2\beta z + \mu^-} = h(g(p_0, m_n), \mu, \mu^-) \quad (B4)$$

where Bayes rule dictates the functions²⁸

$$f: [0,1] \times [0,1] \rightarrow [0,1]$$

$$g: [0,1] \times [0,1] \rightarrow [0,1]$$

$$h: [0,1] \times R^+ \times R^+ \rightarrow [0,1]$$

are given by

$$f(p_0, m) = \frac{\frac{\beta z}{2\beta z + \beta\mu + 2\beta l} p_0}{\frac{\beta z}{2\beta z + \beta\mu + 2\beta l} p_0 + \frac{\beta z + m\beta\mu}{2\beta z + \beta\mu + 2\beta l} (1 - p_0)}$$

$$g(p_0, m) = \frac{\frac{\beta l}{2\beta z + \beta\mu + 2\beta l} p_0}{\frac{\beta l}{2\beta z + \beta\mu + 2\beta l} p_0 + \frac{\beta l + (1 - m)\beta\mu}{2\beta z + \beta\mu + 2\beta l} (1 - p_0)}$$

$$h(p_1, \mu, \mu^-) = \frac{\frac{\beta z + \mu}{2\beta z + \mu} p_1}{\frac{\beta z + \mu}{2\beta z + \mu} p_1 + \frac{\beta z}{2\beta z + \mu^-} (1 - p_1)} \times \frac{\beta z}{2\beta z + \mu^-}$$

It is clear that (one does not even have to differentiate explicitly to conclude the monotonicity)

$$\partial_2 f < 0$$

$$\partial_2 g > 0$$

$$\partial_1 h > 0 \quad \partial_2 h > 0 \quad \partial_3 h < 0$$

We are now ready to prove the lemma. By way of contradiction, assume either

$$m_p(1 - p_0) < m_n(p_0) \quad (*)$$

or

$$m_p(1 - p_0) = m_n(p_0) < 1 \text{ and } \mu^-(p_0, b_2(LS)) < \mu \quad (**)$$

²⁸ The functions f and g also depend on μ . Since f and g do not depend on μ^- , the dependency on μ is of no consequence to our analysis, so we keep it implicit.

We will show that either assumption leads to a contradiction. A contradiction under assumption (*) proves the first part of the lemma; i.e., it proves $m_p(1 - p_0) \geq m_n(p_0)$. A contradiction under assumption (**) proves that if $\mu^-(p_0, b_2(LS)) < \mu$ and $m_n(p_0) < 1$, then $m_p(1 - p_0) \neq m_n(p)$. Together with the proof of the first part (i.e., together with $m_p(1 - p_0) \geq m_n(p_0)$), contradiction under (**) constitutes a proof of the second part of the lemma.

Under either (*) or (**), we have

$$\begin{aligned}
 (1 - a_1(1 - p_0)) &= f(p_0, m_p(1 - p_0)) && \text{by (B1)} \\
 (*)' &\geq f(p_0, m_n(p_0)) && \text{because } \partial_2 f < 0 \\
 &= b_1(p_0) && \text{by (B2)} \\
 &\geq PS(p_0) \frac{\beta z}{2\beta z + \mu^-(b_2(p_0, LS))} && \text{by (6')} \\
 &= h(g(p_0, m_n(p_0)), \mu, \mu^-(b_2(p_0, LS))) && \text{by (B4)} \\
 (**)' &\geq vh(g(p_0, m_n(p_0)), \mu, \mu) && \text{because } \partial_3 h < 0 \\
 (*)' &\geq h(g(p_0, m_p(1 - p_0)), \mu, \mu) && \text{because } \partial_1 h > 0, \partial_2 g > 0 \\
 &\geq h(g(p_0, m_p(1 - p_0)), \mu^-(b_2(p_0, LB)), \mu) && \text{because } \partial_2 h > 0 \\
 &= (1 - PB(1 - p_0)) \frac{\beta z}{2\beta z + \mu} && \text{by (B3)}
 \end{aligned}$$

Now, if assumption (*) holds, then the inequalities denoted with (*)' are both strict. If (**) holds, then the inequality (**)' is strict. Either way, we get

$$(1 - a_1(1 - p_0)) > (1 - PB(1 - p_0)) \frac{\beta z}{2\beta z + \mu}$$

which implies (see equilibrium condition (10')) that $m_p = 1$. This is a contradiction because both (*) and (**) imply $m_p(1 - p_0) < 1$.

Proof of Theorem 1. Under the assumption that $\tilde{p}_0$ is a symmetric random variable, the first part of the theorem follows from the first part of Lemma 3.

For the second part, let p_0 be such that

$$p_0 < \min \left\{ c, \frac{\mu}{z + 2\mu} \right\}$$

This condition holds if the event is sufficiently small, and borrowing costs are sufficiently high. Because $p_0 < c$, Lemma 2 implies that $\mu^-(b_2(p_0, LS)) = \beta\mu$. Because $p_0 < \mu/(z + 2\mu)$, Lemma 2 also implies that $m_n(p_0) < 1$. From the second part of lemma 3, it follows that $m_n(p_0) < m_p(1 - p_0)$. Thus, the probability of observing a sell market order during a small negative event is strictly smaller than the probability of observing a buy market order during a small positive event. This proves the second part of the theorem.

We now consider the third part of the theorem. If $\beta = 1$, then there are no constrained investors, and the model is symmetric. I.e. $m_p(1 - p_0) = m_n(p_0)$. By a continuity argument, it follows that also for β close to one, these probabilities are similar.

In addition, these probabilities are identical whenever constrained investors don't abstain. According to equilibrium condition (14), this is the case whenever the bid in round 2 is greater than the cost of borrowing. This is the case when cost of borrowing is near zero, or p_0 is near one and borrowing cost is less than one; i.e. the event is negative and large.

□

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Table 1: Abnormal returns surrounding unscheduled corporate events

The table reports the descriptive statistics of the cumulative abnormal returns (CAR) surrounding unscheduled corporate events, namely acquisitions, targets, seasoned equity offerings, repurchases, dividend initiations and dividend termination, for a sample of Euronext-Paris stocks in 2003. The sample consists of 95 Euronext-Paris stocks with 101 events. Daily abnormal return for day I is calculated by subtracting the daily index (CAC40) close-to-close returns ($CAC[I-1, I]$) from the daily event's close-to-close returns on day I ($[I-1, I]$). We classify events as positive and negative news based on the CAR observed over Days $[0, +1]$ with Day 0 denoting the event day. Reported are the cross-sectional mean and median for the full sample and the subsamples. In Panel A, we report the CAR over Days $[0, +1]$ while in Panel B, we report the CAR over Days $[-5, +1]$. Panel C reports the CAR ratio which is defined as an event's abnormal returns over Days $[0, +1]$ divided by the event's abnormal returns over Days $[-5, +1]$.

Type of Events	Total	# of Events	Positive		# of Events	Negative	
			Mean	Median		Mean	Median
Panel A: Cumulative abnormal returns [DAYS 0 to +1]							
Overall	101	58	4.84%	2.74%	43	-4.53%	-2.57%
Acquisitions	35	26	3.61%	2.56%	9	-3.88%	-4.15%
Targets	25	16	9.52%	7.12%	9	-7.73%	-2.65%
SEOs	22	8	3.41%	3.42%	14	-4.53%	-3.08%
Repurchases	14	6	2.88%	1.59%	8	-1.83%	-1.82%
Dividend initiations	4	2	2.32%	2.32%	2	-5.34%	-5.34%
Dividend terminations	1	-	-	-	1	-1.82%	-1.82%
Panel B: Cumulative abnormal returns [DAYS -5 to +1]							
Overall	101	58	6.59%	4.25%	43	-4.98%	-3.38%
Acquisitions	35	26	4.81%	3.96%	9	-5.03%	-4.66%
Targets	25	16	12.29%	7.72%	9	-8.10%	-7.35%
SEOs	22	8	4.63%	4.99%	14	-4.63%	-3.67%
Repurchases	14	6	2.07%	1.55%	8	-1.68%	-1.85%
Dividend initiations	4	2	5.66%	5.66%	2	-5.34%	-5.34%
Dividend terminations	1	-	-	-	1	-6.84%	-6.84%
Panel C: CARs ratio of abnormal returns [DAYS 0 to +1]/abnormal returns [DAYS -5 to +1]							
Overall	101	58	44.83%	56.15%	43	73.52%	56.53%

Table 2: Descriptive statistics on sample firms and order flow before corporate events

Panel A reports the descriptive statistics on the Euronext-Paris sample firms associated with five types of unscheduled corporate events, namely acquisitions, targets, seasoned equity offerings, repurchases, dividend initiations and dividend termination. The sample consists of 95 Euronext-Paris stocks with 101 events. We classify events into positive and negative news based on the two day (Day [0,+1]) cumulative abnormal returns with Day 0 denoting the event day. In Panel A, we report market capitalization (in Euros, as of January 2003), daily trading volume (in shares), quoted spread (%age), volatility (%age) and stock price (in Euros) on control Days [-30,-10] before the corporate event. Panel B reports daily statistics on order flow on Days [-5,-1] and control Days [-30,-10] before the corporate event. We calculate the statistics for each firm-event and report the (cross-sectional) statistics across all firm-events. Reported are the daily number of orders and order size for non-marketable as well as marketable orders, the percentage of marketable orders to limit orders and the percentage of limit orders with a hidden size. ***, ** and * indicate that tests of differences are statistically significant at the 1%, 5% and 10% level, respectively.

	Positive events		Negative events	
	Mean	Median	Mean	Median
Panel A: Descriptive statistics on sample firms				
Market capitalization (€ mill)	5,114	425	2,805	227
Daily volume (shares)	454,347	6,034	996,291	45,526
Quoted spreads (%age)	1.32	0.78	1.45	0.88
Daily return volatility (%age)	0.18	0.13	0.25	0.16
Stock price (€)	33.3	27.2	24.1	18.0
	Buy orders		Sell orders	
	Positive events		Negative events	
	Mean	Median	Mean	Median
Panel B: Daily descriptive statistics on order flow				
Daily number of marketable and non-marketable orders				
Days [-5,-1] before corporate event	808.8**	63.9*	995.5*	117.0*
Control period	638.5	50.7	898.6	87.6
Average order size				
Days [-5,-1] before corporate event	1358*	461.7	2031*	1107
Control period	1004	436.6	1771	1045
Percentage marketable\market orders to limit orders				
Days [-5,-1] before corporate event	48.1%***	46.1%***	49.4%	49.0%*
Control period	45.1%	44.3%	50.0%	51.7%
Percentage of limit orders with hidden size				
Days [-5,-1] before corporate event	18.8%	14.8%	17.1%*	16.2%
Control period	18.4%	15.3%	17.6%	16.3%

Table 3: Regression analysis of price aggressiveness and trading costs surrounding unscheduled corporate events

The table reports the regression coefficients that reflect the change in price aggressiveness and trading costs around unscheduled corporate events (Days [-5,+1] with Day 0 denoting the event day) after controlling for order attributes and market conditions. The sample consists of 95 Euronext-Paris stocks with 101 events. We separately investigate buy (sell) orders around positive (negative) events. Price aggressiveness is defined as an ordinal variable that takes the value of 1 when *the order* is classified as the most aggressive category and takes a value of 7 when it is classified in the least aggressive category. Detailed definitions of the categories and all the explanatory variables are provided in Section 3.2. Columns (1) and (2) present coefficients of the price aggressiveness regressions while columns (3) - (8) present the coefficients of implementation shortfall, effective spreads and opportunity costs regressions, following the approach proposed by Perold (1988). For a buy order, *effective spread cost* is defined as the difference between the filled price of each submitted order and the mid-quote price at the time of order submission, *opportunity cost* as the difference between the closing price on the day of order cancellation or expiration and the quote midpoint at the time of order submission, and *implementation shortfall* as the summation of the two costs. For *effective spread cost*, we report regression results conditional on partial execution (*effective spread cost* $\neq 0$, columns 5 and 6). For *opportunity cost*, we report regression results conditional on partial non-execution (*opportunity cost* $\neq 0$, columns 7 and 8). Panel A.1 reports the regression coefficients of market conditions and limit order book variables. Panel A.2. reports individual day coefficients on event Days [-5,+1] that reflects the change in price aggressiveness and trading costs before the event for the same firm relative to control days. Panel B reports cumulative effects that sum up the individual day coefficients on Days [-5,-1]) relative to control period (Days [-30,-10]) for the same firm. Time series coefficients are estimated on an event-by-event basis. Reported results are aggregated across events using the Bayesian framework of DuMouchel (1994) described in Appendix A. We also report the t-statistic of the difference between positive and negative events.

Variable	Price aggressiveness		Implementation shortfall		Effective spread cost: fill rate >0%		Opportunity cost: fill rate < 100%	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient (1)	Coefficient (2)	Coefficient (3)	Coefficient (4)	Coefficient (5)	Coefficient (6)	Coefficient (7)	Coefficient (8)
Panel A.1: Regression coefficients of market conditions and limit order book variables								
Intercept (t-statistic)	4.8866 (79.72)	4.8747 (63.23)	0.0541 (3.03)	0.0560 (2.81)	0.0651 (5.77)	0.0466 (4.57)	0.0969 (2.21)	0.0775 (2.59)
Price aggressiveness (t-statistic)			-0.1933 (-0.80)	0.0005 (0.43)	22.4275 (6.05)	19.8713 (4.09)	-0.0039 (-0.72)	0.0005 (0.44)
Order exposure (t-statistic)	0.6876 (13.90)	0.5768 (9.24)	-0.0175 (-2.95)	-0.0034 (-0.69)	-0.0236 (-6.15)	-0.0066 (-4.17)	-0.0255 (-2.93)	0.0211 (3.28)
Total order size (norm) (t-statistic)	-0.0030 (-2.87)	0.0004 (0.46)						
Order size (million shares) (t-statistic)			0.0584 (0.83)	-5.3622 (-0.11)	-5.9155 (-0.13)	-0.0592 (-1.99)	-0.5211 (-0.80)	2.6222 (0.01)
Bid-ask spread (norm) (t-statistic)	0.0252 (2.43)	0.1116 (3.27)						
Depth -same side (norm) (t-statistic)	-0.0226 (-4.83)	-0.0420 (-6.56)						
Depth - opposite side (norm) (t-statistic)	0.0034 (1.86)	0.0147 (2.25)						
Book order imbalance (norm) (t-statistic)	-0.0008 (-1.15)	-0.0009 (-1.17)						
Volatility (norm) (t-statistic)	0.0039 (0.68)	0.0016 (0.11)	0.0004 (0.19)	-0.0029 (-2.10)	0.0046 (2.22)	0.0049 (3.63)	0.0015 (0.59)	-0.0055 (-2.55)
Waiting time (norm) (t-statistic)	0.0176 (2.82)	0.0138 (1.49)						
Trade frequency (norm) (t-statistic)	-0.0612 (-2.58)	-0.0531 (-2.26)	-0.0035 (-1.55)	0.0051 (1.76)	-0.0003 (-0.37)	0.0014 (1.81)	-0.0128 (-3.01)	0.0057 (1.49)
Lag (price aggressiveness) (t-statistic)	-8.9688 (-9.18)	-6.1587 (-4.59)						
Lag (displayed order size (norm)) (t-statistic)	-0.0040 (-1.54)	0.0062 (1.88)						
Last trade size (norm) (t-statistic)	0.0020 (2.21)	0.0059 (2.38)						
HiddenOppSide (norm) (t-statistic)	-0.0395 (-7.25)	-0.1111 (-5.61)						
First trading hour indicator (t-statistic)	0.0028 (0.10)	0.0813 (1.75)						
Last trading hour indicator (t-statistic)	-0.1489 (-5.79)	-0.1785 (-8.03)						
Market volatility (norm) (t-statistic)	0.0004 (0.16)	-0.0005 (-0.49)						
Industry volatility (norm) (t-statistic)	0.0010 (1.51)	0.0001 (0.08)						

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Variable	Price aggressiveness		Implementation shortfall		Effective spread cost: fill rate >0%		Opportunity cost: fill rate < 100%	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A.2: Individual Day Minus 5 to Day Minus 1 Day Dummies								
Day Minus 5 (dummy) (t-statistic)	-0.1689 (-1.46)	0.1423 (2.23)	0.0275 (0.88)	-0.0553 (-1.42)	0.0044 (0.39)	-0.0004 (-0.11)	0.0491 (1.14)	-0.1210 (-2.13)
Day Minus 4 (dummy) (t-statistic)	-0.0711 (-0.98)	0.0494 (1.01)	0.0037 (0.12)	-0.0115 (-0.42)	-0.0055 (-1.23)	0.0032 (0.50)	-0.0197 (-0.34)	-0.0406 (-0.74)
Day Minus 3 (dummy) (t-statistic)	-0.1032 (-2.04)	0.0138 (0.24)	0.0846 (2.67)	-0.0350 (-2.59)	-0.0006 (-0.10)	-0.0026 (-0.68)	0.1338 (2.23)	-0.0691 (-3.14)
Day Minus 2 (dummy) (t-statistic)	-0.1501 (-1.69)	-0.0034 (-0.07)	0.0161 (0.36)	0.0114 (0.33)	0.0029 (0.21)	0.0092 (1.18)	0.0008 (0.01)	-0.0152 (-0.28)
Day Minus 1 (dummy) (t-statistic)	-0.1088 (-2.61)	0.0475 (0.54)	0.1018 (2.61)	-0.0142 (-0.48)	0.0327 (2.14)	-0.0080 (-1.38)	0.1146 (1.56)	0.0005 (0.01)
Day 0 & Plus 1 (dummy) (t-statistic)	-0.0309 (-0.91)	0.0324 (0.56)	0.0830 (3.51)	-0.0181 (-0.78)	0.0005 (0.13)	-0.0042 (-1.68)	0.1292 (3.39)	-0.0231 (-0.54)
Day Plus 2 (dummy) (t-statistic)	-0.0080 (-0.17)	0.0215 (0.27)	0.0011 (0.05)	-0.0381 (-1.67)	0.0043 (0.74)	0.0001 (0.02)	0.0162 (0.39)	-0.0647 (-1.84)
Panel B: Cumulative Effect of Day Minus 5 to Day Minus 1 Coefficients								
Cumulative Effect: Day Minus 5 to Day Minus 1 (t-statistic)	-0.5617 (-2.71)	0.2488 (1.65)	0.2422 (2.43)	-0.0771 (-1.10)	0.0107 (0.50)	0.0020 (0.19)	0.3234 (2.17)	-0.1849 (-1.77)
T-statistic of the Difference between Positive and Negative Events	-3.16		2.62		0.37		2.79	

Table 4: The impact of short sale constraints on informed trader strategies and the stock price drift

The table reports the percentage of marketable to limit orders, the number of trades, the change in price aggressiveness and the drift in stock price (i.e., opportunity cost of a limit order) surrounding unscheduled corporate events. Panels A.1. and B.1. report the percentage of marketable to limit orders on Days [-5,-1] and in the control period (Days [-30,-10]) for the same firm. Panels A.2. and B.2. report the percentage change in the number of trades on Days[-5,-1] from the control period. Panels A.3. and B.3. present the cumulative effect measure on Days [-5,-1] relative to the control period based on a regression analysis of order that controls for order attributes, market conditions and limit order book. Detailed definitions of price aggressiveness and the explanatory variables are provided in Table 3 and Section 3.2. Panels A.4. and B.4. present the cumulative effect measure on Days [-5,-1] relative to control days based on a regression of opportunity cost of limit orders that controls for order attributes and market conditions. Opportunity cost is defined in Table 3 and Section 3.2. We investigate buy (sell) orders around positive (negative) events based on whether or not the stock (i) belongs to SBF 120 index (Panel A, columns (1) to (4)), (ii) is eligible for Euronext's SRD-facility (Panel A, columns (5) to (8)), (iii) has exchange traded options (Panel B, columns (1) to (4)), and (iv) has event period absolute CAR above 5 percent (Panel B, columns (5) to (8)). For Panels A.3., A.4, B.3 and B.4, the time series coefficients are estimated on an event-by-event basis. Reported results are aggregated across events using the Bayesian framework of DuMouchel (1994). For Panels A.1. and B.1. we test for significant differences between Days [-5,-1] and the control period. For Panels A.2. and B.2. we test for significance of the percentage change in the number of trades from the control period. ***, ** and * indicate that tests are statistically significant at the 1%, 5% and 10% level, respectively. Lastly for all panels we also report the t-statistic of the difference for each panel's measure between positive and negative events.

Panel A	Events From Companies in SBF120		Events From Companies not in SBF120		Events From SDR-Eligible Companies		Events From SDR-Ineligible Companies	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of corporate events	24	16	34	27	25	24	33	19
Panel A.1.: Percentage of marketable\market orders to limit orders								
Days [-5,-1] before corporate event	48.4%*	57.1%**	47.7%**	47.4%**	48.4%**	56.7%**	47.9%*	46.2%***
Control period	46.3%	54.7%	44.1%	51.5%	44.9%	54.8%	45.2%	55.6%
T-statistic of the difference between positive and negative events		-0.13		2.44		1.08		2.87
Panel A.2.: Number of trades by type of Event								
Days [-5,-1]: % change from control period	21.3%***	26.4%***	22.4%***	11.9%***	13.6%***	24.5%***	32.4%***	9.6%***
Control period	919.1	975.0	20.9	199.8	848.7	848.4	59.7	50.4
T-statistic of the difference between positive and negative events		-0.47		1.74		-1.29		2.32
Panel A.3.: Cumulative effect of Day Minus 5 to Day Minus 1 coefficients based on price aggressiveness regression analysis								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.3583 (-2.19)	-0.6229 (-1.63)	-0.8417 (-2.07)	0.3451 (1.86)	-0.4245 (-2.12)	-0.5309 (-2.07)	-0.8976 (-2.03)	0.4242 (2.28)
T-statistic of the difference between positive and negative events		0.64		-2.66		0.33		-2.75
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Panel A.4.: Cumulative effect of Day Minus 5 to Day Minus 1 coefficients based on opportunity cost regression analysis								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	0.4327 (2.25)	0.2075 (2.08)	0.4180 (2.21)	-0.2347 (-1.72)	0.3627 (1.92)	0.1303 (1.60)	0.4407 (2.15)	-0.3547 (-1.58)
T-statistic of the difference between positive and negative events		1.04		2.78		1.13		2.62
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes

Panel B	Events From Companies with Options		Options		Event Period Absolute Return > 5%		Event Period Absolute Return < 5%	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of corporate events	14	15	44	28	19	13	39	30
Panel B.1.: Percentage of marketable\market orders to limit orders								
Days [-5,-1] before corporate event	50.2%*	66.1%***	47.5%**	39.6%**	53.5%***	51.7%	46.6%	49.4%**
Control period	48.5%	62.2%	44.0%	45.3%	43.7%	50.8%	45.5%	52.8%
T-statistic of the difference between positive and negative events		-1.67		3.04		2.12		1.97
Panel B.2.: Number of trades by type of Event								
Days [-5,-1]: % change from control period	29.4%***	22.3%***	18.1%***	8.7%***	25.4%***	22.3%***	12.7%***	8.9%***
Control period	1293.1	1266.3	106.0	62.7	228.6	746.7	509.9	452.7
T-statistic of the difference between positive and negative events		0.66		1.81		0.43		0.98
Panel B.3.: Cumulative effect of Day Minus 5 to Day Minus 1 coefficients based on price aggressiveness regression analysis								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.2338 (-2.12)	-0.8409 (-2.10)	-0.7233 (-2.35)	0.4368 (2.36)	-1.4365 (-2.09)	-0.5929 (-1.83)	-0.3301 (-2.01)	0.2479 (1.92)
T-statistic of the difference between positive and negative events		1.46		-3.23		-1.11		-2.76
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Panel B.4.: Cumulative effect of Day Minus 5 to Day Minus 1 coefficients based on opportunity cost regression analysis								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	0.2456 (1.54)	-0.2082 (-0.77)	0.4511 (2.76)	-0.2052 (-1.44)	0.4696 (2.44)	0.1457 (1.01)	0.3928 (2.10)	-0.3711 (-2.21)
T-statistic of the difference between positive and negative events		1.45		3.02		1.35		3.04
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes

Table 5: Robustness analysis of the impact of short sale constraints on informed trader strategies

The table reports robustness analysis of informed traders strategies before unscheduled corporate events. The table reports the change in order aggressiveness observed on Days [-5,-1] relative to control days [-30,-10] from a regression analysis of price aggressiveness on order attributes, market conditions and limit order book. We report results for buy (sell) orders around positive (negative) events based on whether or not the stock (i) belongs to SBF 120 index (Panel A, columns (1) to (4)), (ii) is eligible for Euronext's SRD-facility (Panel A, columns (5) to (8)), (iii) has exchange traded options (Panel B, columns (1) to (4)), and (iv) has event period absolute CAR above 5% (Panel B, columns (5) to (8)). Panels A.1 and B.1 report regression coefficients based on a specification with a single dummy coefficient for Day[-5,-1]. Time series coefficients are estimated on an event-by-event basis and aggregated using the Bayesian framework of DuMouchel (1994). Panels A.2. and B.2. present the cumulative effect measure based on individual Days [-5,-1] dummy coefficient in a panel regression specification that is estimated separately for each of the 16 event type. The specifications include event fixed effects and clustered standard error by date. Panels A.3. and B.3. present the cumulative effect measure based on individual Days [-5,-1] dummy coefficient in a panel regression specification that is estimated separately for samples (i) to (iv). Within each panel specification, the regression coefficients for subsamples of positive and negative events are obtained using interaction term variables. The specifications include event fixed effects and clustered standard error by date. Lastly, for all panels we also report the t-statistic of the difference for each panel's measure between positive and negative events.

Panel A	Events From Companies in SBF120		Events From Companies not in SBF120		Events From SDR-Eligible Companies		Events From SDR-Ineligible Companies	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of corporate events	24	16	34	27	25	24	33	19
Robustness A.1.: Single Dummy Coefficient for Days[-5,-1] - Individual Event Regression Aggregated based on Panayides (2007) and Bessembinder et al. (2009)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.0664 (-2.48)	-0.1293 (-1.86)	-0.1066 (-2.43)	0.0972 (2.30)	-0.0636 (-1.81)	-0.1047 (-2.22)	-0.0845 (-2.10)	0.1312 (2.43)
T-statistic of the difference between positive and negative events	0.84		-3.35		0.70		-3.21	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Robustness A.2.: Cumulative Effect of Day Minus 5 to Day Minus 1 Coefficients - Panel Regression by Event Type (16 types)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.2084 (-1.91)	-0.8778 (-4.30)	-0.8202 (-3.63)	0.4264 (1.87)	-0.2636 (-2.50)	-0.6696 (-4.31)	-0.6026 (-2.23)	0.6872 (1.92)
T-statistic of the difference between positive and negative events	2.89		-3.88		2.15		-2.88	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Robustness A.3.: Cumulative Effect of Day Minus 5 to Day Minus 1 Coefficients - Pooled Regression aggregated by Short Sale measure (4 measures)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.2177 (-2.09)	-0.9576 (-4.44)	-0.9251 (-4.21)	0.4066 (1.92)	-0.2762 (-2.84)	-0.6560 (-4.24)	-0.4051 (-1.89)	0.5539 (2.48)
T-statistic of the difference between positive and negative events	3.29		-4.28		2.20		-2.95	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes

Panel B	Events From Companies with Options		Options		Event Period Absolute Return > 5%		Event Period Absolute Return < 5%	
	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events	Buy orders, Positive events	Sell orders, Negative events
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of corporate events	14	15	44	28	19	13	39	30
Robustness B.1.: Single Dummy Coefficient for Days[-5,-1] - Individual Event Regression Aggregated based on Panayides (2007) and Bessembinder et al. (2009)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.0540 (-2.37)	-0.1263 (-1.39)	-0.0946 (-2.38)	0.0887 (2.50)	-0.1071 (-1.70)	-0.0950 (-1.32)	-0.0612 (-2.05)	0.0374 (1.37)
T-statistic of the difference between positive and negative events	0.77		-3.44		-0.13		-2.44	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Robustness B.2.: Cumulative Effect of Day Minus 5 to Day Minus 1 Coefficients - Panel Regression by Event Type (16 types)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.2382 (-1.97)	-0.6910 (-3.99)	-0.7563 (-4.81)	0.5313 (2.28)	-0.4895 (-2.05)	-0.7428 (-3.53)	-0.2430 (-2.83)	0.2203 (1.02)
T-statistic of the difference between positive and negative events	2.15		-4.58		0.79		-2.02	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes
Robustness B.3.: Cumulative Effect of Day Minus 5 to Day Minus 1 Coefficients - Pooled Regression aggregated by Short Sale measure (4 measures)								
Cumulative Effect:								
Day Minus 5 to Day Minus 1 (t-statistic)	-0.2317 (-2.15)	-0.7290 (-4.29)	-0.7981 (-5.14)	0.3662 (1.99)	-0.5624 (-2.72)	-0.6778 (-3.58)	-0.1936 (-1.97)	0.3170 (1.59)
T-statistic of the difference between positive and negative events	2.60		-4.93		0.41		-2.34	
Control variables in regressions	yes	yes	yes	yes	yes	yes	yes	yes

Fig. 2: The speed of learning before positive and negative events

Fig. 2 plots the speed of price discovery before corporate events based on the coefficients from an “unbiasedness regression” specification, following Biais, Hillion and Spatt (1999). For each half-hour time period (i) in the interval Days $[-5,-1]$, we regress the close-to-close return (ret_{cc}) in interval Days $[-6,+1]$ on return from the close on Day $[-6]$ to the end of time period i (ret_{ci}). For both positive and negative events, we estimate the unbiasedness regressions cross-sectionally for each time period. The figure plots the beta coefficient and the confidence intervals (dotted line) based on standard error of the beta coefficient. The bar graph below shows the ratio of the root mean square error (RMSE) of each unbiasedness regression to the unconditional standard deviation of the close-to-close return (ret_{cc}).

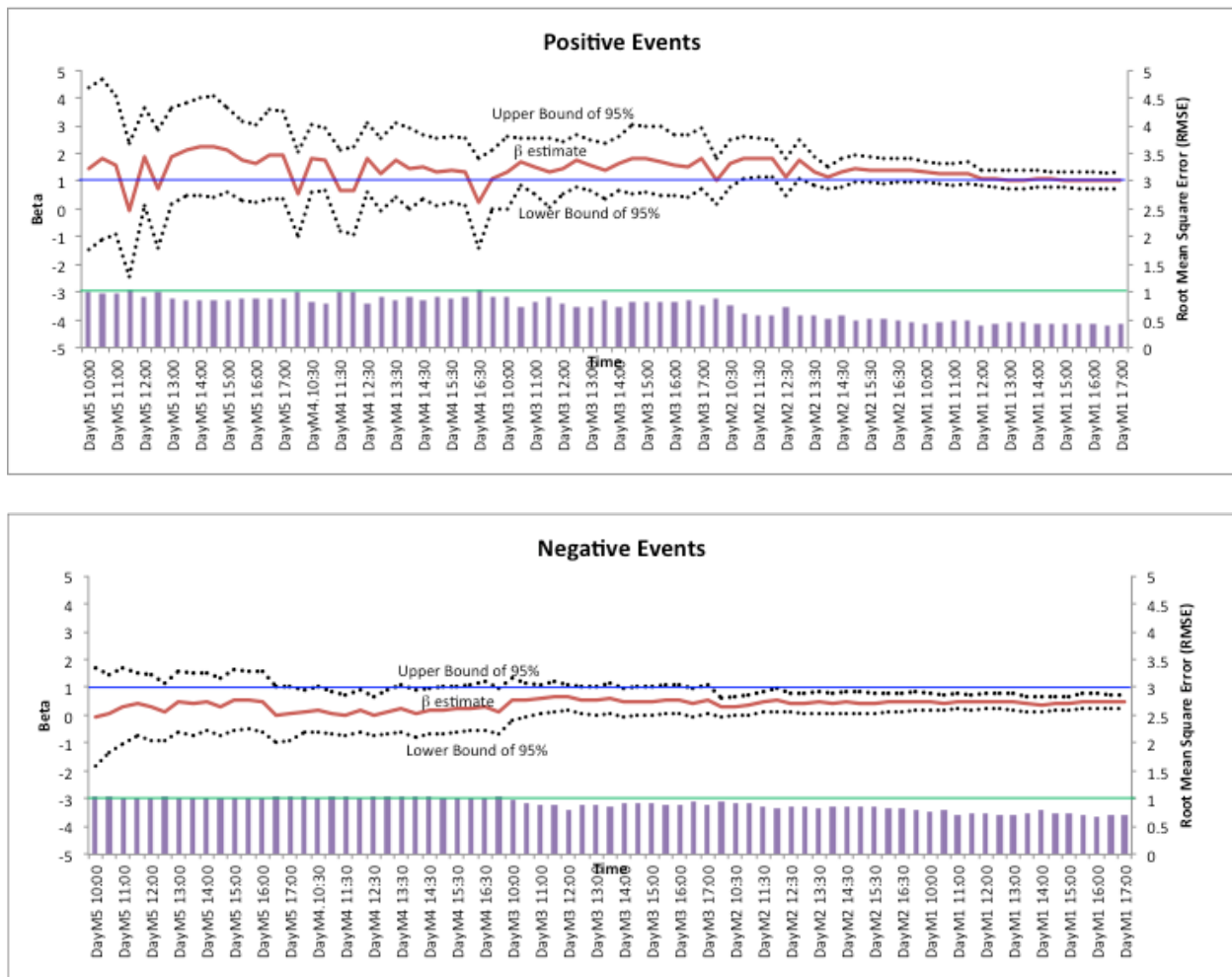


Fig. 3: Short sale constraints and the speed of learning before corporate events

Fig. 3 plots the speed of price discovery before corporate events based on the coefficients from an “unbiasedness regression” specification, following Biais, Hillion and Spatt (1999). For each half-hour time period (i) in the interval Days $[-5, -1]$, we regress the close-to-close return (ret_{cc}) in interval Days $[-6, +1]$ on return from the close on Day $[-6]$ to the end of time period i (ret_{ci}). The figure plots the beta coefficient and the confidence intervals (dotted line) based on standard error of the beta coefficient. The bar graph below shows the ratio of the root mean square error (RMSE) of each unbiasedness regression to the unconditional standard deviation of the close-to-close return (ret_{cc}). The model predicts that informed traders use less aggressive order in short constrained stocks before negative events. The proxy for short sale constraint is eligibility for Euronext's SRD facility.

